A Six-Month ECDLP Research Programme: Implementations, Experiments, Structural Leads, and Evidence

A comprehensive retrospective of the aburan28 research portfolio, 19 April–19 October 2026

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19 October 2026

Abstract

This paper reviews six months of work across the `aburan28` cryptographic research portfolio, centred on `crypto`, `cryptanalysis`, `crypto-autoresearcher`, and `ml-cryptanalysis`. Nineteen pinned repository snapshots contain 305,367 tracked entries. Their reachable histories contain 34,276 commits in the review window; metadata across 5,144 current branch heads contains 49,936 distinct commits. These are measures of recorded activity, including merges, imports and upstream history, not counts of discoveries. A companion catalogue indexes 4,075 research, experiment, benchmark and manuscript directories and 16,587 safely parsed ledger files.

The technical record spans prime fields, ordinary binary curves, Koblitz curves, small extension fields, higher-genus transfers, summation polynomials, Boolean and prime-field algebraic solvers, isogeny graphs, endomorphisms, GPU arithmetic, learning, and formal verification. Concrete progress includes substantially cheaper decomposition and elimination stages, complete toy pipelines, modest held-out improvements to production IC, a 2.645-second cold-process saving in an N83 holdout, and reproducible structural-discovery experiments. A downloaded isogeny checkpoint contains 46,960 attempts at 2,935 audited vertices; all 29,409 archive-manifest entries pass a fresh byte check. Nineteen frontier sessions have 3,178 verified measured executions and 54 exhausted executions, with matching record hashes. These checks are distinguished from fresh cryptographic replay.

The record establishes engineering and methodological progress. It does not establish a general prime-field ECDLP algorithm below rho, a cryptographic-scale index-calculus break, or a proof of intrinsic hardness variation across an isogeny class. Stage results, retained-precomputation online results, complete cold solves, finite observations, conditional arguments and checked-proof claims are kept separate. Open branches are reported independently, including large-memory PKM runs and an Semaev defect-impact audit. The result is an evidence map, an assessment of leads, and a set of explicit next proof and measurement obligations.

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1 Scope, evidence, and the meaning of progress

The purpose of this retrospective is to explain what was built, which questions were tested, what improved, what was corrected, and what remains unresolved. Its organising question is the elliptic-curve discrete logarithm problem (ECDLP): given a public point G of order r and $Q = [d]G$, recover $d \bmod r$. The programme approaches this question from several directions at once: faster generic solvers, non-generic relation generation, algebraic systems, alternative representations and groups, structural prediction, and reproducibility infrastructure.

The repositories contain enough work to require both a readable synthesis and a searchable catalogue. This paper supplies the synthesis; the accompanying CSV and JSON files supply the catalogue. It extends the earlier fourteen-week index-calculus manuscript to April–October and to the wider portfolio. The earlier manuscript is a useful guide, but its figures and conclusions are re-read against their source studies and later corrections rather than inherited wholesale.[PRIOR]

1.1 Corpus and temporal boundary

The review window is 19 April through 19 October 2026, UTC. The main `crypto` repository was created on 25 April; consequently this window contains its entire initial development. Snapshots were captured independently on 19 October. They are not an atomic snapshot of all machines, branches and external storage. The `cryptanalysis` account default was an active development branch; the primary content review deliberately pins `main` separately. This matters: its `main` snapshot has 48,811 tracked entries, while the initially observed default branch had 16,091.

Repository	Entries	Window commits	Pinned commit
WDSat	89	2	`cf146c780f2e`
autolab	305	0	`f1c783082328`
cairn	920	537	`6bb3a52c88f5`
cryptanalysis	48,811	963	`6e170e48882e`
crypto	114,791	8,499	`4f9ea9480b17`
crypto-autoresearcher	138,314	22,794	`d0f70c0a0905`
crypto-lean	33	4	`148f5dc389b3`
cryptologic-archives	156	3	`1096f83076a7`
ecbench	9	1	`5ca79764b725`
ecc-challenges	433	101	`4b4d47349b53`
ecdlp-cost-challenge	79	49	`2e215f1d503e`
ecdlp-prime-field-smallness	66	5	`4f20be9b36ff`
ecdsafail-challenge	50	1,172	`b6b648f4377f`
eigenx	201	12	`88f88a2da7bc`
ghs-trapdoor	2	3	`93981a150972`
isogeny-labs	136	38	`d613806737c7`
ml-cryptanalysis	214	71	`9c06ddee943c`
rho-blackwell	320	18	`5c3db3907924`
the-sqisign	438	4	`4c42deee64ee`

"Entries" counts the tracked tree, not source lines, distinct programs, or experiments. "Window commits" counts unique commits reachable from that snapshot, filtered on committer time. It includes merge commits and the history of imported or forked code. For example, the 1,172 window commits reachable from the quantum-challenge fork cannot be credited as 1,172 contributions by this programme. Conversely, the Autolab snapshot's zero window commits does not mean that no Autolab experiments occurred: results and modified work are present in other branches and in the autoresearcher migration records.

The three central histories contain 32,256 window commits: 8,499 in `crypto`, 963 in `cryptanalysis/main`, and 22,794 in `crypto-autoresearcher`. The portfolio's 34,276 snapshot-reachable commits are distinct by commit hash across these nineteen snapshots. Current branch metadata extends the union to 49,936 commits over 5,144 heads. Deleted branches, unreachable commits, uncommitted local work and unavailable external artifacts remain outside that union. Branch content is selectively reviewed; fetching its commit metadata is not a scientific audit of every branch.

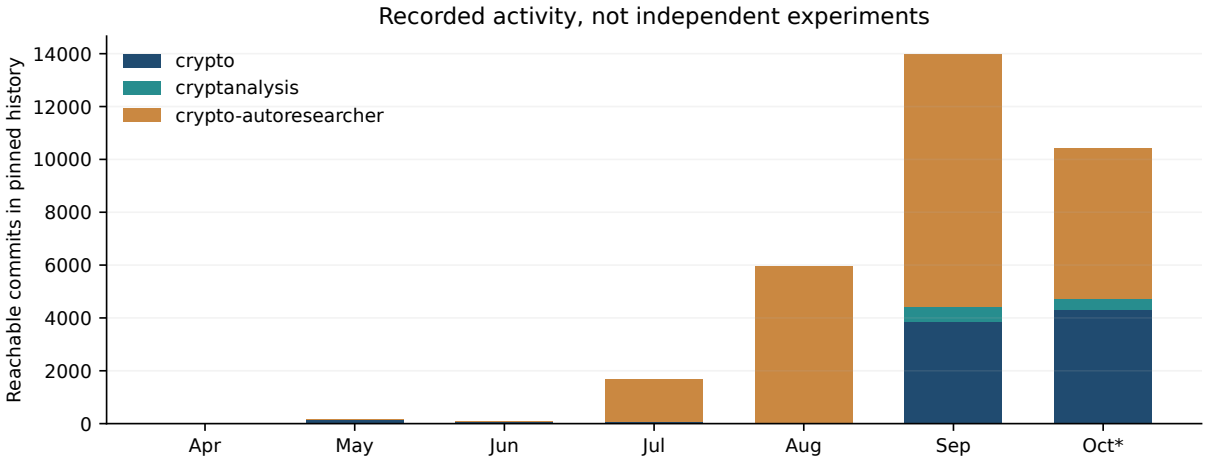


Figure 1: Recorded activity in the three core pinned histories by committer month. Merge and non-merge commits both occur in the histories. Counts indicate archival volume, not independent experiments, researcher hours, authorship, novelty or scientific success. October is partial.

1.2 How the review was carried out

The collection procedure used blobless Git histories, full tracked-tree inventories, sparse content materialisation, repository indexes, ledger parsing, and focused reads of studies and their code paths. The catalogue contains 21,006 materialised Markdown documents under the documented size and directory policy. A document's presence in that index means its title, path and bytes were catalogued; it does not mean that every sentence was manually assessed. A directory is similarly a navigational unit: several directories can describe one study, while one directory can contain many runs.

The 4,075 directory records cover `research/`, `experiments/`, `benchmarks/` and `papers/` across the snapshots. The ledger catalogue parses 16,587 YAML files with a safe loader; 24 additional files could not be parsed by that loader and are retained as explicit failures. Some root-level legacy records, archives and checkpoints repeat or supersede other records. Record totals therefore cannot be interpreted as distinct hypotheses or independent validations. Both source paths and identifiers are retained so a reader can deduplicate for a particular question.

The review freshly checked byte hashes of all 19 frontier-session record files, counted their outcomes, checked a downloaded volcano archive against its complete manifest, and built the native C library and covered its 17 configured tests. Existing reports supply most experimental measurements. The review did not rerun the full historical corpus, rent accelerators, rebuild every Rust target, or rebuild Lean dependencies. Every result below retains the distinction between source inspection, fresh metadata verification, historical solver replay, and a new solver execution.

1.3 Access limits are part of the record

The historical Autolab migration reports 186,032 files and 51,286,418,925 bytes across 32 task packages. Its 48-GB-class raw mirror is a local working-tree archive, with manifests and materialisation receipts in Git; it was not available as raw bytes in this runtime. Its structured port comprises 84 historical EXP/RUN packages: 32 ALPF, 11 ALBIN, 38 ALISO and three ALECF. These are archival claims with their original scope, not freshly replayed results.[MIGRATION]

Likewise, the ten-lead advisory register preserves hashes and statements for private source packets without including those packets. The public register can be inspected, but full artifact replay cannot be inferred from its structural validity. In contrast, the linked 37,926,124-byte volcano checkpoint was accessible and was downloaded and checked. These are different access situations and are reported differently.[TEN] [VOL-ARCHIVE]

1.4 A multidimensional assessment

Progress has at least five observable dimensions: implemented capability, verified correctness, lower cost on a fixed task, stronger experimental instruments, and narrower uncertainty about a mechanism. The owner's current criterion explicitly treats a verified improvement over the current IC baseline as success, including constant-factor engineering. Beating rho is a longer-term objective, not the admission criterion for every useful implementation change.[C-RULES]

This paper accordingly preserves engineering gains even where IC remains slower than rho. It also preserves negative results and corrections as knowledge, while not counting them as speed improvements. A review verdict such as "ADVANCES" in a study's local vocabulary is translated into its actual boundary: a constant-factor improvement, a fitted exponent on a bounded panel, a primitive saving, or an accounting correction.

The volume supports describing a substantial, highly instrumented research programme. It does not support ranking the effort against all human ECDLP research: comparable records of unpublished work, labour, compute and independent scientific output are unavailable. This paper therefore measures the accessible programme rather than asserting a historical rank.

1.5 Development over six months

The first-parent history provides a useful chronology when interpreted as provenance rather than proof. On 25 April, the initial educational library was followed by known-answer tests and hardening work. Early May added curve arithmetic, a broad attack taxonomy, binary curves, pairings, summation-polynomial IC, cryptanalysis tooling and research benches. The same period explored canonical lifts, Smart-style adaptations to non-anomalous curves, p-adic constructions, coordinate correlations, GLV/HNP ideas and geometric transfers. Several commit subjects use strong phrases such as "structural obstruction"; later revisions show why those subjects must not be treated as permanent conclusions.

May–June Autolab work tested Howe gluing, Richelot and CM patterns, GLV-related lattices, separator hypotheses and numerical behaviour. The sequence includes a concrete example of progress through falsification: successive proposed predictors of a lattice effect failed on expanded cross-curve panels. The October LLL correction then withdrew a claimed curve-specific failure after tracing it to floating-point behaviour, including a diagnosis already present in the May numerical note. [AUTOLOG][HNP][LLL]

July introduced the dedicated autoresearcher framework; the late-July and August work imported historical experiments, built ledger and coordination contracts, and expanded the research corpus. September concentrated on Koblitz IC, SAT/F4 comparisons, invariant and compact-orbit bases, native arithmetic and the independent cryptanalysis toolkit. October added ecbench, more explicit cost accounting, root-finding oracles, field and genus ladders, learning and symbolic discovery, large graph checkpoints, equal-advice controls and shared Lean infrastructure. These stages overlap: they describe development emphasis, not a sequence in which one research direction replaced all the others.

2 Mathematical setting and the field taxonomy

The size of a field, the order of the whole curve group, and the order of the subgroup containing the logarithm must remain separate. Throughout, q denotes a field cardinality, r a prime subgroup order, h a cofactor, and $Q = [d]G$ a target in $\langle G \rangle$. A curve over an 83-bit field can contain a much smaller prime subgroup. Its ECDLP is priced in that subgroup, not in the field size.

2.1 Generic reference algorithms

Pollard rho obtains a collision among combinations $[a]G + [b]Q$ and solves the resulting scalar congruence. The familiar random-collision normalisation is approximately $\sqrt{\pi r/2}$ group steps under idealised assumptions. This is a model constant, not a universal implementation bound. Negation, efficiently computable automorphisms, fruitless cycles, distinguished-point density and restart policy change observed constants. Parallel collision search changes throughput and coordination costs without by itself changing the square-root exponent.[6, 8]

Baby-step giant-step spends square-root memory to obtain another generic tradeoff. Kangaroo is appropriate when the unknown scalar is known to lie in an interval; its scale is the interval width, not

automatically r. Pohlig–Hellman decomposes composite orders, so a "large curve" whose largest prime factor is small is not a large prime-order ECDLP instance. These distinctions are implemented in the native toolkit and in the challenge infrastructure.[CA-README][COST]

Generic-group lower bounds establish asymptotic limitations under an explicit oracle model. They should not be read as a sharp leading-constant claim of exactly $\sqrt{r/2}$ operations for every implementation with free negation. The cost challenge is valuable as an executable metering instrument; this retrospective does not adopt every explanatory constant in its README as a separately proved lower bound.

2.2 Index calculus as a complete method

Choose a factor base $\mathcal{B} = \{F_1, \ldots, F_B\}$, or a quotient of it under known symmetries. A relation expresses a sampled point as a short sum of base points. For example,

$$[a]G + [b]Q = \sum_i c_i F_i \quad \Longrightarrow \quad a + bd \equiv \sum_i c_i \log_G(F_i) \pmod{r}.$$

After enough independent relations, linear algebra and a target descent recover the logarithm. Relation count, matrix rank, and recovered scalars are separate checks. A large number of valid but dependent rows can leave the system unable to determine the requested target.

For a random-set heuristic with B points and m unordered summands, the expected number of decompositions is roughly $B^m/(m!r)$. This is not a universal yield theorem: repeated points, signs, orbit stabilisers, cofactors, projection into $\langle G \rangle$ and collisions can change the count. The programme repeatedly found that exact subgroup and base conventions matter.[BOUNDARY][HEUR]

Semaev's summation polynomials encode the existence of point sums through their abscissae. They provide algebraic formulations, but replacing a group problem by a polynomial system does not make the system cheap. The cost of constructing, solving, lifting and verifying that system is central to the programme.[7]

Table 2: Field and problem regimes in the portfolio. Evidence in one row does not automatically transfer to another.

Regime	Structure used	Principal limitation in this record
$E(\mathbb{F}_p)$	coordinate subsets, multiplicative towers, low-degree maps, special endomorphisms	no general sub-rho prime-field IC demonstrated
Ordinary $E(\mathbb{F}_{2^n})$	$\mathbb{F}_2$ subspaces, Weil restriction, Boolean algebra	decomposition degree and total yield remain limiting
Koblitz $E/\mathbb{F}_2$ over $\mathbb{F}_{2^n}$	Frobenius orbits and known subgroup eigenvalues	rho receives the same available symmetry; setup is material
Composite-degree binary fields	proper subfields and GHS descent	weak classes, genus and explicit maps determine applicability
Prime-degree binary extensions	no proper intermediate subfield	composite-degree GHS conclusions require a new construction
$E(\mathbb{F}_{p^k})$, small fixed k	base-field abscissae, Gaudry/Joux–Vitse, torsion invariants	expensive algebraic constants and linear algebra
Higher-genus Jacobians	divisor arithmetic, covers, large primes	source-subgroup transport and same-problem reference must be verified
Supersingular isogeny problems	graph navigation, quaternion and Kani constructions	an isogeny problem or KEM is not the ordinary ECDLP

2.3 Prime, binary, and extension fields

In a prime field, Frobenius exists but is the identity on field elements; there is no proper intermediate subfield or nontrivial $\mathbb{F}_p$-linear coordinate subspace. Saying that "$\mathbb{F}_p$ has no Frobenius" is therefore an imprecise shorthand. The actual obstacle is the absence of a useful nontrivial subfield decomposition of the kind available over extensions. This does not rule out nonlinear sparse sets or other non-generic algorithms.

In $\mathbb{F}_{2^n}$, an $\mathbb{F}_2$-subspace of dimension ℓ supplies a compact membership condition for a factor base. Weil restriction turns an extension-field polynomial into Boolean equations. The behaviour of the resulting ideal depends on the formulation and product spaces, not merely the nominal field degree. On Koblitz curves,

the map $\pi : (x, y) \mapsto (x^2, y^2)$ is also a curve endomorphism. Its action on the selected cyclic subgroup can be represented by a known eigenvalue $\lambda \bmod r$, allowing orbit-weighted relations.[3]

Over $\mathbb{F}_{p^k}$, a base defined by $x \in \mathbb{F}_p$ interacts with summation polynomials through Weil restriction. Fixed-degree extension-field algorithms and their symmetry reductions are established antecedents of the implemented work. Their asymptotic opportunities do not imply reasonable constants at the measured sizes, nor do they imply an algorithm on $E(\mathbb{F}_p)$.[1, 2, 4]

The implemented studies therefore cover prime fields, binary extensions, and selected odd-characteristic extensions, notably degrees two through six, together with protocol-specific extension arithmetic. This is not a claim of a complete backend for every finite field, every characteristic or every elliptic-curve model. Backend limits and unimplemented field degrees remain separate from mathematical applicability.

2.4 Isogenies and higher-dimensional transfers

An isogeny defined over the working field is a homomorphism on rational points. If its restriction to the prime subgroup is injective, then $\varphi(Q) = [d]\varphi(G)$ preserves the logarithm. A destination solver must still solve that transported instance more cheaply after charging map discovery, construction, evaluation and recovery. A small degree, shared j-invariant, common group order or common endomorphism field is not by itself an executable transport certificate.

A cover or Jacobian route has more obligations: a concrete destination, an explicit nonzero map, a proof or check of subgroup preservation, an implementable destination solver, and the total source-to-answer cost. Merely finding a low-genus curve with desirable arithmetic does not establish that the source ECDLP can reach it. The portfolio's newer identity and curve registries explicitly encode these distinctions. [TRANSFER][TEN][VOLCANO]

3 How the experiments measure progress

The most consequential methodological change is the movement from a headline counter to a complete, explicit accounting boundary. The same research direction can look favourable when only relation probes are counted, unfavourable when each probe is priced, and different again when setup and linear algebra are included. The programme's rules were written in response to concrete examples of this failure mode.[C-RULES] [RESIDUAL]

3.1 Cost boundaries and comparable units

The historical normalisation is

$$S = \frac{\text{total charged group-addition-equivalent work}}{\sqrt{r}}.$$

Within an explicit curve, subgroup and implementation, this makes a square-root generic reference roughly flat. It does not make different primitive weights, memory envelopes or target laws interchangeable. Some studies count field multiplications, some Boolean word XORs, some simulated instructions, and some admitted wall time. A conversion must be measured or explicitly modelled before units can be joined.

A cold solve includes reusable setup, factor-base construction, indexes, relation collection, matrix work, target work and verification. An online solve starts after reusable state has been prepared and excludes that state's construction. Both questions are useful. The online question requires a comparator with the same advice and reuse opportunity; the cold question requires setup on both arms. A ratio between IC cold process time and rho online time can be recorded as a diagnostic, but is not a matched cold method ratio.

The current one-target contract also separates target-dependent phases. It disallows replacing a one-target rho reference with a multi-target batch, or dividing a shared collision search by target count and calling the result one-target latency. Repeated one-target workloads form a panel; they do not become a single batch ECDLP simply because the statistics are computed together.[C-RULES][N83]

3.2 Unpriced work and the sensitivity band

Many historical counters omit native operations. Rho can omit canonicalisations, partition hashes and table operations; IC can omit root calls, native matrix work or membership tests. If $L_A \le C_A$ and $L_B \le C_B$ are incomplete cost lower bounds, then L_A/L_B does not bound C_A/C_B in either direction. A ratio of lower bounds is not a lower bound on the ratio.

The October accounting rule therefore prices emitted unpriced operations at several common weights, including $w \in \{0, 0.1, 1\}$ GAE, on all arms. A ranking is accepted only if it survives the band. A missing counter emission cannot be repaired by choosing a weight: it remains an implementation gap. A pinned primitive calibration can supplement the band but cannot silently replace it on one side only.[C-RULES]

The band is a sensitivity analysis, not a claim that a 64-bit XOR and a hash-table lookup really have identical cost. This is especially visible for F4: weighting every XOR at 0.1 GAE is much more expensive than one study's pinned calibration near 0.002 GAE. The conservative conclusion must state where it depends on the band or on a particular calibration. [FRONTIER]

3.3 From exploratory scripts to sealed records

The record includes exploratory scripts, toy known-answer ladders, Autolab tournaments, immutable experiment contracts, ICMS and ecbench sessions, native compiler and emulator checks, hardware experiments, and independent replay receipts. The ecbench instruments add explicit specifications, method and workload identities, per-run outcomes, source and binary hashes, host manifests, cost phases, replay audits and comparison records. The SQLite index is a view of evidence, not a substitute for the sealed evidence.

Earlier scripts remain useful historical observations. They do not automatically meet newer isolation, primary-online-window or replay contracts. In particular, the migration of a result into a dashboard cannot improve its evidence tier. The research orchestrator likewise separates an executor's observation from a coordinator's official status transition and an independent review.[AR-README][C-RULES]

3.4 Controls and statistical design

Several recurring practices are worth retaining: freeze the protocol before the measurement; tune on disjoint training points; test on held-out public points; interleave arms; include identical A/A controls; verify every returned scalar or root set; retain failures; and bootstrap over the actual independent workload or curve unit. Five targets are not 25 independent targets merely because five configurations ran on each. Similarly, repeat-bootstrap intervals on one scalar panel do not become coverage over arbitrary scalar distributions.

Positive controls show that an instrument can detect the proposed kind of signal; null controls show whether the same apparent signal arises without the mechanism. The ML studies use class-held-out splits and label permutation; decomposition studies compare with exhaustive point sums; GPU studies compare ranks and pinned variables across paths. A failed control is retained even when the main result appears favourable.

Resource failures are a separate outcome. Exhaustion means the declared search budget ended. A timeout or memory cap says something about the specific execution, not nonexistence of the desired relation. A failed artifact-integrity gate can make a run inadmissible without supplying negative evidence about the mathematics.

3.5 A freshly checked campaign

The 15 October frontier campaign is a stronger common reference than a collection of unrelated best cells. Its 19 sessions cover eight Koblitz curves, six NIST-shaped prime curves and five prime Kummer-ladder curves, with 16 public targets, two measured rounds and a warm-up. The sessions contain 4,848 executions: 4,769 verified and 79 exhausted including warm-ups. Removing warm-ups leaves 3,232 measured executions: 3,178 verified and 54 exhausted. Every stored record-file hash matches its session manifest in this review. The stored audits contain 3,178 successful identical replays of verified measured records; exhausted records are retained separately, not reported as successful replays. [FRONTIER][BAND]

No IC arm in that campaign beats the cheapest matched constant-memory rho arm at any of the three band weights. Compact-orbit IC is closest on the Koblitz ladder: its IC/rho ratio is 3.8–15.0 with the extra

native work priced at zero, and 4.9–17.8 at $w = 0.1$. Prime pair-table IC grows from approximately 10.5 to 101.8 times rho at $w = 0.1$ over the 20–40-bit ladder. The campaign's fitted exponent differences are not uniformly separated at every weight. This establishes a bounded comparison on those implementations and instances, not a theorem that all future IC must lose.

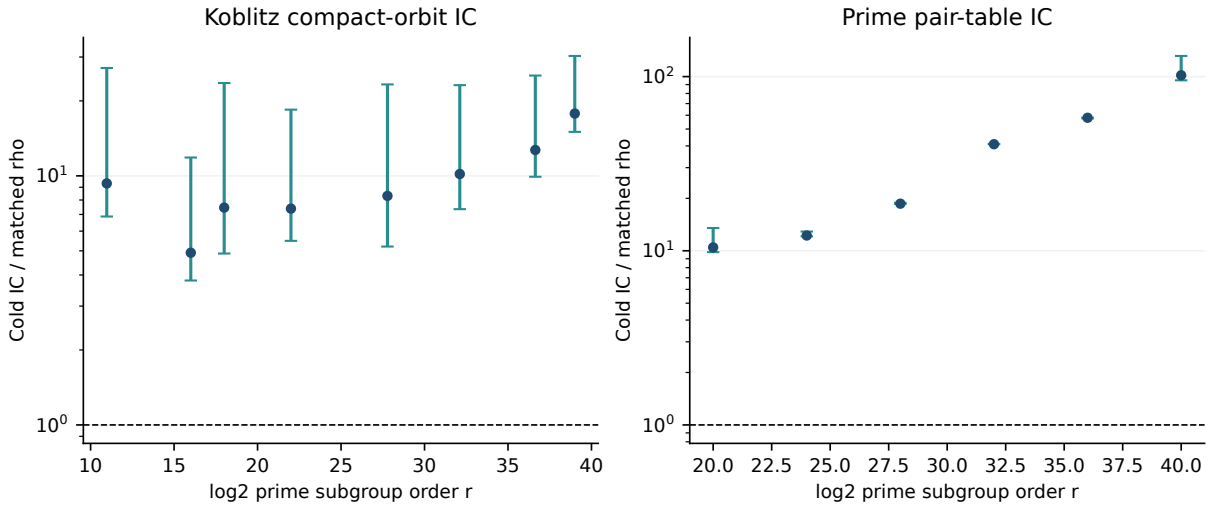


Figure 2: The campaign's recorded cold IC/rho ratios. The centre marks $w = 0.1$; the spans show $w = 0$ through $w = 1$. Every displayed cell has 32 verified measured runs. These are counted-work comparisons at L0, not new native wall-time measurements. Koblitz points are ordered by subgroup size, which need not follow field degree. Source: [BAND].

The campaign does not supersede every later experiment. Its role is a fixed, comparatively well-accounted baseline. Later improvements require their own matched comparison before being used to move this frontier.

3.6 Corrections as an experimental capability

The programme now has repeated examples of a result being made less favourable by better accounting: a relation-only crossover disappearing after linear algebra was priced, F6 geometry dominating its saved XORs, and online IC losing an advantage after rho was given preprocessing. These are scientific corrections. They are also evidence that the instruments can reject a preferred interpretation. They should remain in the record beside the successful optimisations rather than being removed from a progress narrative.

4 Prime-field research: sparse sets, root finding, and algebraic bases

Prime fields are the most demanding generalisation target in this portfolio. Coordinate arithmetic is efficient, but the subfield conditions used by extension-field IC are absent. The programme tests several substitutes rather than treating this absence as a proof that no alternative can exist.[PRIME]

4.1 What the smallness experiments actually learned

The dedicated smallness repository asks whether an inexpensive coordinate-defined set behaves additively better than a matched random set. Its experiments include intervals in x and y, slopes, affine and Möbius images, Hamming weight, Farey heights, multiplicative subgroups, random windows, Fourier spectra, low-degree polynomial sweeps, Coppersmith constructions, Witt carries and learned log-bit predictors. The main closure panel uses three roughly 20-bit prime-order toy curves. [SMALL]

An illustrative observation is especially clear. For a base near size 2,048, ordinary coordinate predicates have pooled two-sum closure ratios near one: $x < 2048$ gives about 1.04, a subgroup predicate about 0.95, and matched random sets about 0.98–1.00. A log-interval positive control, which uses unavailable discrete-log information to define the set, gives 256.4 against a predicted 256.5. The control shows that the

instrument can see strong additive structure; the measured cheap predicates do not show that structure on the tested panel.

The null comparison must match sign closure as well as cardinality. Negation-closed sets have correlated pair sums, so a naive Poisson error bar understates their spread. Random-window maxima must also be compared with maxima from the same-size null search. The lesson is a concrete experimental discipline, not a universal theorem that every inexpensive predicate is random.

The Witt-carry and small-root studies examine why tempting analogues of extension coordinates become expensive. Their polynomial degrees and success thresholds belong to the specific constructions tested. A failed learned log-bit predictor is similarly evidence about the models, features, labels and curve sizes used. It does not prove that no computational representation of prime-field structure can help.

4.2 A useful reduction of the two-summand oracle

For a short-Weierstrass model $y^2 = x^3 + ax + b$ in characteristic other than two or three, and affine points R, F with $x_R \neq x_F$, the familiar chord formulas yield

$$S_3(x_R, x_F, X) = (x_R - x_F)^2\big(X - x(R+F)\big)\big(X - x(R-F)\big).$$

Thus, once F is fixed, the roots needed for two-summand decomposition are already group-law expressions. One can subtract F and test base membership, handling signs and exceptional points explicitly, rather than compute a fresh Gröbner basis. The proof source formalises the affine algebraic identity; its coordinate definitions and hypotheses are visible in the Lean file.[RF-PROOF]

This is an implementation opportunity, not a new general summation-law theorem or a prime-field ECDLP breakthrough. The source audit explicitly identifies prior sum-and-membership algorithms. It also corrects the manuscript's broad lower-bound language: a reusable pair table can make a repeated query constant-time after quadratic setup, so an unqualified linear-per-query lower bound would be false.[RF-AUDIT]

The implementation family includes batched inversions for direct prime subtraction, binary pair-sum tables, three- and four-summand meet-in-the-middle variants, windowed scans, dense root indexes, Kummer-base variants and a CUDA root-finding kernel. The prime and binary models use their own group laws. The short-Weierstrass identity should not be silently substituted for the generalized binary model $y^2 + xy = x^3 + ax^2 + b$.[RF-PDP] [RF-GPU]

4.3 Measured improvements, with the comparator repaired

The source audit reports binary per-decomposition ratios of 383 and 1,535 against matrix F4 at field degrees nine and thirteen. The precomputed pair-table build is excluded from those query ratios. At degree 23 a complete binary solve is reported 122 times faster with the pair table, while F4 is faster at degrees five, nine and thirteen. This reversal is evidence that setup and scale matter.[RF-AUDIT]

On five NIST P-curves the six-point-base query ratios are 3,021–5,802 against the benchmark's Buchberger implementation. The printed benchmark label said "F4", but the timed source calls `groebner_f4::buchberger`. The corrected comparator belongs in every reuse of those ratios. These experiments perform tiny-base oracle work on large field parameters; they do not solve an unknown NIST-curve logarithm.

The complete prime-DLP comparison against msolve has two evidence tiers. At 16 bits, both totals are directly measured. At 20 and 24 bits, the F4 totals are composed from sampled per-query CPU cost and the root-finder's measured query count. The reported factors of about 100, 105 and 50 must not all be described as directly timed complete solves. Nor are any of them comparisons against the strongest generic rho method.

The ecbench audit supplies a different view. On one registered 20-bit prime subgroup, a pair table reduces the counted IC S from 1,524.511 to 28.753 relative to direct subtraction. Its paired counted ratio is 0.018861. Native work remains unpriced on those prime arms, so that quotient does not bound a complete-cost ratio. On a degree-17 Koblitz instance, the corresponding IC comparison is fully priced in that study's calibration: 356.894 → 20.360, paired ratio 0.057047. Even there, the rho comparison has unpriced work and cannot be promoted into a complete-cost speed claim.[RF-AUDIT]

4.4 Kummer towers and the remaining algebraic question

Petit–Kosters–Messeng-style bases use multiplicative and tower structure to give summation systems more algebraic organisation. [5] In the recorded five-curve ecbench root-finding panel, 896 runs verify and the IC/rho ratios remain about 29–141. Changing a plain base to a tower base of the same size changes that table-based oracle's cost by roughly one percent. The conclusion is narrow: the tower's algebraic structure is not exploited by that particular enumeration/table mechanism. It says nothing decisive about a genuinely algebraic tower solver.[PKM]

The large-memory work now addresses that distinction. At the pinned main snapshot, several larger tower cells remain bounded or resource-limited. An open branch subsequently records the $m = 3, N = 21$ run on a 124-GB, 32-vCPU host. Both targets finish all degree-seven reductions without a refutation, then stop at the first degree-eight step under the frozen dense-cap rule. The demanded B' has 79,330,205,871 entries, approximately 317 GB at four bytes per entry; the recorded peak resident memory is 67.1 GB. The two runs take 6,445 and 6,416 seconds.[PR2160]

This is real execution progress relative to an unrun memory-gated card. It does not supply a completed degree or a productive degree-eight refutation. The branch's results explicitly leave the interpretation between the protocol's inconclusive and lower-bound cases to the coordinator. It would be inaccurate either to keep calling the cell "never run" or to call it a successful degree-eight solve.

Another open branch integrates prime tower F4, signature F5, an F6-style closure and enumeration into ecbench. It reports 140 verified runs on four 9–15-bit prime-order curves. F5 reduces the emitted F4 work on the same trees, while F6's closures save essentially no total work. The author-reported priced comparisons still favour enumeration and rho. The first attempt is explicitly voided for thread-dependent replay. These are unmerged engineering observations at their pinned PR head, not an update silently applied to main's frontier.[PR2223]

4.5 Earlier prime-curve structural work

The May–August record also includes p-adic lifts, anomalous-curve techniques applied outside their usual setting, CM analysis, GLV/HNP lattices, twists and coordinate correlations. Its value includes identifying which proposed bridge was absent. An efficient scalar decomposition, a low-height coordinate or a numerical lattice anomaly does not automatically yield an efficient unknown-scalar recovery algorithm. The later LLL correction makes this especially tangible: apparent differences between named curves can be implementation artefacts.[HNP][LLL][STRUCTURAL]

5 Ordinary binary and Koblitz research

5.1 What the Koblitz structure buys

The Frobenius-invariant base implementation checks invariance point by point, folds columns into Frobenius orbits and carries the corresponding known λ^k weights into relations. Small-field records report roughly degree-n savings in relation bookkeeping and degree-n^2 savings in linear algebra, consistent with the known invariant-base method. Rho also uses the available signed-Frobenius quotient, whose idealised improvement is of square-root orbit size. Comparing IC to rho without that available quotient would credit IC with a symmetry the reference was unnecessarily denied.[KOBLITZ][3]

The progression from invariant subspaces to divisor bases, SAT-optimised bases, rank-guided constructions and compact signed-Frobenius orbit bases is an important implementation record. The compact base stores orbit representatives and their known scalar weights rather than an entire quadratic pair table. It trades memory and rank-generation behaviour against online probes. It is the closest IC family to rho in several recorded small-curve comparisons, but its advantage relative to another IC construction is not a claim that it beats matched rho.[COMPACT] [FRONTIER]

5.2 Production improvements that survive a narrow reading

Free relations harvested from the root table are a concrete example of a mechanism that is useful but smaller than first hoped. A degree-61 census with 600 columns finds 147 nontrivial collisions but only 49 independent relations against 599 required. The original predicted yield was about 60 times too high. That

defeats the proposed large rank shortcut at the tested construction; it does not make every free relation worthless. [EQUAL]

The later production comparison tunes the baseline and candidate rather than comparing a tuned candidate to an untuned incumbent. Its held-out candidate/incumbent counted ratio is 0.960, with interval [0.946,0.976], over 360 verified runs. The feature is on by default with a switch restoring the previous path, and the study records exact A/A behaviour. At the same fixed K, the degree-41 and degree-53 intervals include one; degree 61's 0.962 remains the clean same-K effect. The earlier pooled 0.957 result is annotated not to be reused as the tuned claim.[ROUND1][ROUND2]

This is measurable, modest production engineering. It also illustrates why "a major lead did not produce its hoped-for yield" and "a useful constant-factor improvement landed" can both be true.

5.3 N83: an exact cold-process example

The five-point N83 holdout compares nearby compact-base sizes under a frozen winner rule and Tier-1 cloud conditions. Its curve is `icv1-f2m83-t6151469093347-9eba6cd5`, with model $y^2 + xy = x^3 + x^2 + 1$, prime subgroup order $r = 8,569,786,107,849,059$ and cofactor 1,128,547,018. The subgroup has about 53 bits; "N83" is the field degree, not an 83-bit prime-order subgroup.[N83][N83-ROWS]

Table 3: Recorded five-point cold-process means. The baseline's two A/A child times are averaged per point before the panel mean. This is a configuration comparison within IC, not a comparison of matched cold IC and rho. Source: [N83-ROWS].

Columns	Usable points	Cold seconds	Setup seconds	Rank seconds
1,104	183,264	120.227	48.728	71.320
1,108	183,928	121.296	48.955	72.171
1,120	185,920	119.826	50.081	69.525
1,152	191,232	122.471	53.067	69.255

The 1,120-column arm saves 2.645 seconds over 1,152 columns, with paired 95% interval [2.085,3.205] seconds. The 1,104-column arm saves 2.244 seconds, interval [1.980,2.508]. But the observed 0.401-second advantage of 1,120 over 1,104 columns has interval [−0.236,1.038], so the frozen selection remains `BOUNDED_UNKNOWN` with no selected candidate identifier. All 25 IC arms and five rho arms have recorded independent replay, and the study retains five passing baseline A/A gates.

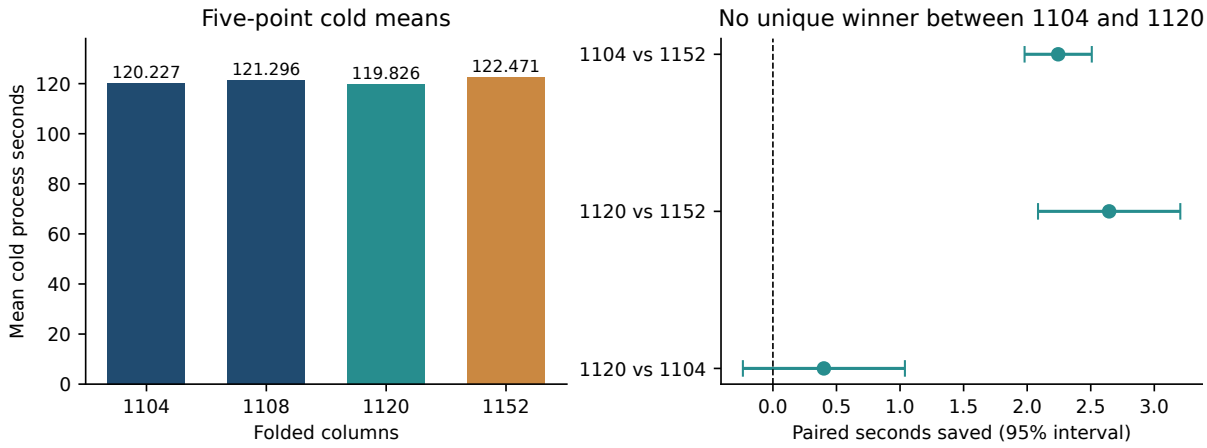


Figure 3: A verified, held-out IC configuration improvement without an overstated winner. Left: recorded cold-process means. Right: paired seconds saved over the 1,152-column baseline, with the recorded 95% intervals; the final row compares the two leading candidates and includes zero. Five public workloads, Tier-1 cloud instance. Source: [N83-ROWS].

The setup/rank split identifies the next bottleneck more usefully than a headline online ratio. Rank generation accounts for about 69–72 seconds in these means, while target online work is a fraction of a second. Reducing retained-state query latency alone cannot remove the cold rank cost. Different base sizes also move the two phases in different directions, which is why the winner rule should remain attached.

5.4 Online ladders and the preprocessing control

The historical ladder reaches degrees 53, 61, 71, 73 and 83, including 128-bit field packing beyond degree 64. The degree-61 two-host panel records a 13.94 online rho/IC median ratio with interval [9.81,19.13] on 64 fresh workloads, with roughly 28 seconds of setup excluded. These are actual observations under that retained-preparation question. [PRIOR][COMPACT]

The equal-advice audit changes their interpretation. At degrees 41, 53 and 61, preprocessing rho uses the same or less retained advice and is reported 53–317 times faster online by ratios of means; setup costs are separately shown and are not matched in every such row. Some controls fail their original prediction, and the host is contended. Consequently the retrospective preserves the measurements, their control caveats and their setup boundary, rather than treating them as a universal dominance theorem.[EQUAL]

At degree 83, a later study runs time-matched and RSS-matched Bernstein–Lange controls on 16 public points. Its K3300 time-matched median IC/preprocessed-rho ratio is 3.74, interval [2.65,11.05]; its RSS-matched ratio is 40.61, interval [24.27,71.72]. Its K600 comparisons are larger. Under that protocol, neither original "IC beats rho" online interpretation survives matched preprocessing. The study also states that no operation-count ranking is admitted: one band's order flips, while another arm omits root-call emission.[EQUAL83]

The equal-bytes, single-process-group wide experiment at degrees 73 and 83 is nevertheless a distinct open requirement. Its instrument is built and checked, but the frozen wide session is explicitly not run. Time-matched and RSS-matched experiments should not be relabelled as that missing experiment.[WIDE]

5.5 Ordinary binary oracles and solving degree

The ordinary-binary programme includes XOR-aware SAT, Semaev systems, matrix F4, inherited F4, pair-and-solve oracles, meet-in-the-middle tables and root finding. A binary MITM ladder reaches $\ell = 12$, where the recorded table has about 8.09 million sums and occupies about 259 MB. The maximum counted query work is 16,092 point additions against about 8.39 million quartic solves in the comparator; witnesses and negative answers are cross-checked on the panel. The table is target-independent setup and is charged as such. The large query improvement is not an end-to-end exponent improvement.[TIMELINE]

The degree-closure studies make another important distinction: plain XL degree, first-fall degree, closure refutation degree and the cost of a full Gröbner solve are not interchangeable. Full closure can refute a system several degrees below an XL-only diagnostic. In the recorded scale studies, two explanations survive the tested cells: a rise near $\ell = 7$, or a rise when $m\ell \geq 21$. The $m = 4, \ell = 6$ cell separates them and remains unmeasured at the required boundary. The sparse closure is an independent correctness check on smaller instances, but is too slow to resolve that boundary in the recorded experiment.[DREG]

For ECC2K-130, degree-five extrapolations are therefore optimistic scenarios rather than measured cryptographic-scale solvers. The reported $2^{71.13}$ row is in unconverted column-touch units, whereas the $2^{60.81}$ rho figure is in group operations. A displayed "10.32-bit gap" is an arithmetic comparison of those model entries, not a calibrated attack-cost ratio. Neither a memory failure nor the absence of a favourable row proves a general lower bound on binary IC. [DREG][LEADS18]

6 Algebraic engines, SAT, and linear algebra

The solver programme is broader than a single F4 implementation. It includes Boolean matrix F4, signature/F5 variants, inherited elimination over assignment trees, geometry-assisted F6-IC, prime-field F4, tower arithmetic, sparse and dense kernels, Four-Russians elimination, SAT/XOR encodings, linear-row harvesting, GPU offload and destination matrix solvers. These are reusable instruments, even where the current ECDLP pipeline prefers an enumerative oracle.[GB][CA-README]

6.1 Inherited elimination and the accounting lesson

Inherited F4 carries a parent's reduced information into a child partial assignment rather than constructing a fresh Macaulay matrix at every node. On the identical-tree K13 and random-13 panels, the record reports matrix-work reductions of 198.6 and 222.1 relative to matrix F4. F5's word-XOR ratios against F4 are about 0.6465 and 0.6414. These are specific stage counters on particular systems, not a claim that every F4 task is hundreds of times faster.[GB][PRIOR]

F6-IC adds factor-base domain pruning and meet-in-the-middle leaves. Its word-XOR count is lower than inherited F4, and its relation stage can be faster. Once the leaf geometry and batched additions are charged, however, the older complete-cost study reports F6-IC costs 64.6–68.8 times inherited F4 at degree 13 and 1,303–1,353 times at degree 19. The saved algebraic counter moved work into another phase. That is an accounting loss on those workloads, not a reason to delete the stage measurement or to assume every future geometry-assisted engine must lose.

The engine audit also found a prime-field soundness bug, contract defects and latent holes. Its regression and brute-force comparisons are part of the progress: a fast engine that can discard a real root is not an acceptable oracle. Correctness must cover both positive witnesses and negative decisions, not just the final answer on favourable targets.

6.2 What the external comparisons establish

The prime-field F4 work reports 29.8–54.1 improvements over its predecessor on square generic systems, with msolve/ours ratios 0.96–1.22 in the recorded single-thread comparison. Reduced bases are checked against external tools. This is a useful example of making the external baseline explicit. It is not a comprehensive fastest-solver claim over other degrees, fields, orderings, thread counts or hardware. [FP-F4]

Task identity is equally important. Deciding whether a Boolean system has an admissible root, extracting a few linear consequences, and computing a complete reduced basis are different jobs. The fact that a specialised oracle decides a target in milliseconds while a general CAS spends seconds producing a full basis does not by itself establish a corresponding engine speedup on equal outputs. The portfolio's newer external-comparison rule requires the same inputs, outputs and host, and names missing external tools rather than selecting a convenient slower reference.[C-RULES][PRIOR]

6.3 GPU results: real acceleration and a negative deployment result

The RTX 5090 experiment is particularly informative because it records both. A degree-six Macaulay task with 168,363 rows and 188,203 columns takes 425.8 seconds for host build plus elimination and 5.70 seconds for device build plus elimination, a recorded factor of 74.7 on that pod. Another degree-six matrix improves 25.5 times. Ranks, refutations and pinned variables agree. The device column includes first-use CUDA/NVRTC cost in the first elimination; separate upload and kernel times are also reported.[F4-GPU]

Yet the small per-node device decider does not beat the production inherited host engine. On its fixed 1,024-target oracle panel, the GPU path takes 5.57 seconds while the default inherited host path takes 1.25 seconds. The GPU is useful on the large matrices; the better host algorithm wins on the small repeated systems. A proposed deployment should use this result to decide where to offload, rather than project the large-matrix factor onto the whole IC pipeline.

The H100 follow-up records a kept tiled update, 1.11–1.49 times faster on the measured matrices, and a reverted merged-basis panel that is 1.5–5.3 times slower. Upload costs remain. This is a concrete record of an optimisation search with a retained unsuccessful arm.[F4-PANEL]

The root-finding CUDA kernel is a separate case: the cited implementation is host-emulated, with no GPU measurement in that study. Its validation cannot inherit the F4 kernel's hardware evidence merely because both use CUDA.[RF-GPU]

6.4 Factor-base geometry and empirical solver selection

The cryptanalysis product-profile study measures 656 factor bases from seven families over nine fields. For two-summand systems it fits the empirical rule

$$e = n - \dim V - \dim V^{(2)}, \qquad D = 2 + [e < -3.5] + [e < -8.5],$$

with mean absolute error 0.06 degrees, or 0.08 when leaving out each field size. This is a fitted rule on that dataset, not a proved degree law for growing parameters. In selected cells, geometric progression bases reduce query cost 15–40 times relative to random bases while leaving yield similar.[HEUR]

The three-summand comparison shows why formulation is another research lever. Direct coordinates and symmetric variables live in different product spaces. Minimising the dimensions of $V^{(k)}$ can reduce the

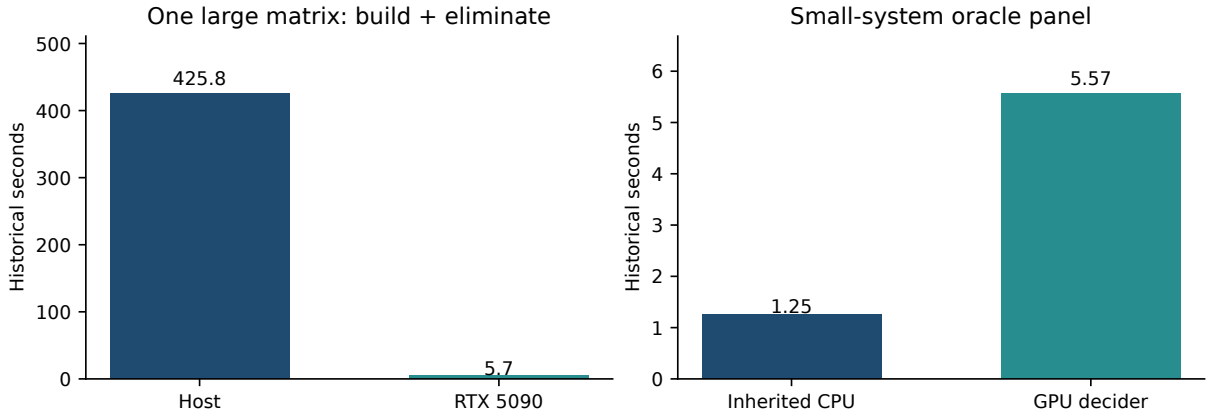


Figure 4: Two different boundaries in the RTX 5090 study. Left: one large degree-six matrix, host and device build plus elimination. Right: the small-system oracle panel, where the inherited CPU engine is faster than the GPU decider. Historical wall observations, not a new measurement or an end-to-end ECDLP speedup. Source: [F4-GPU].

number of symmetric unknowns even at the same solving degree. The report explicitly labels these rows PDP-stage profiles and keeps their candidate identity unset until a calibrated complete pipeline exists.

It also identifies a practical stopping condition: a relation collector can wait forever for full rank when the attainable matrix rank is one less than the number of columns. Solving modulo the known kernel and verifying the target is different from assuming that another dependent row will repair it. This is a concrete lesson about implementations, not only asymptotic analysis.

6.5 Linear algebra and harvested information

Linear algebra is the phase most likely to invalidate a relation-only crossover. The residual-walk record documents a claimed relation-phase crossing that disappears after matrix cost is included. Sparse filtering, structured Gaussian elimination, Lanczos, Wiedemann and block Wiedemann are therefore method components, not cleanup after the interesting part. [RESIDUAL]

The later genus-three experiments compare complete solvers after changing large-prime handling and matrix strategy. One structured-elimination and Wiedemann route has recorded complete candidate/incumbent ratio 0.653 [0.647,0.660] at $q = 2^{20}$, with 474 verified logs in its panel. Its linear-algebra constant improves while the fitted exponent of that phase rises. A subsequent block-Wiedemann change gives about 0.945 at 2^{21}; a combined schedule and solver reaches 0.605 of the earlier round-one baseline. These factors belong to their named comparison chains and are not multiplied into an invented cumulative factor on the prime-field ECDLP.[ROUND2]

Linear-row harvesting is another modest, testable gain. The recorded root-set panel has zero failures on 96 comparisons; selected oracle cells improve about 1.07–1.17 times. The report also states where the bench gain fails to carry into production because the production solve stops at its first accepted root. The exact stopping policy is part of the workload definition.

7 Small extension fields and higher-genus pipelines

This work is where the programme most clearly explores known non-generic asymptotic routes rather than trying to manufacture prime-field structure. The results are interesting precisely because complete costs can be measured against the same problem. They remain separate from deployed prime-field curves and prime-degree binary challenges.

7.1 The residual-walk and parity ledgers

The $\mathbb{F}_{p^3}$ harness studies subspace factor bases, residual walks, large primes, GLV symmetries, decomposition oracles and sparse matrices. Its corrections show that an oracle that reduces the number of walked

residuals can make the method more expensive if the per-residual work increases. One cited example reduces a headline walked count by around two orders of magnitude while total cost worsens by about 4.6 times. [RESIDUAL]

The same record prices linear algebra that earlier rounds omitted. The linear-algebra growth then dominates a relation-phase exponent and changes the claimed crossover. This provides an unusually concrete negative result: the bottleneck is named and priced, rather than the route being declared unsuccessful because it did not immediately beat rho. The parity ledger extends this discipline to several degrees and decomposition choices.[PARITY]

7.2 Degree four: symmetry improves a real algebraic constant

On suitable curves over $\mathbb{F}_{p^4}$, a rational two-torsion point and an appropriate coordinate make the summation system invariant under a D_4 action. The unweighted invariant formulation reduces a generic 4,096-dimensional quotient to 512. Its recorded decomposition constant at $p = 269$ is about $2.54 \cdot 10^{10}$ base-field multiplications, approximately 48 times below the plain formulation.[PRIOR]

The weighted four-variable follow-up is stronger evidence of iteration than quoting that original factor forever. It replaces the five-variable presentation by invariants of weights $(2, 2, 2, 1)$ and records a 3,979-row, 4,200-column matrix rather than the previous 13,412 by 11,628 matrix. At $p = 269, 1033, 4093$, median counted costs are around $1.4 \cdot 10^9$ multiplications including lift-back and reruns at the next degree when required. This is roughly another 18 times below the previous invariant formulation, and about 844–863 times below the named plain reference at those primes.[D4]

The study keeps exceptions. At $p = 17$, one of six planted targets is degenerate; the other five agree with brute force and MITM. The nine generic positive-control systems have the expected quotient dimension. The gain is a decomposition-stage constant. The study explicitly does not measure end-to-end S for that weighted formulation.

Its crossover calculations are therefore model outputs. Earlier models moved a degree-four handover from around 2^{150} to 2^{135}; the weighted follow-up reports a further handover near $2^{118.9}$ in its particular model. Those are not three observed ECDLP crossings. They depend on the relation rate, cofactor, matrix model, unit conversion and the constant measured at much smaller sizes.[D4][LEADS18]

The subsequent complete-method experiments also show that combining individually attractive ideas need not help. A D4-plus-double-large-prime variant makes linear algebra cheaper but total cost worse on its held-out panel. Stopping at the first actual dependency then repairs part of that loss: the later complete candidate/incumbent ratio is 0.979 [0.973,0.986] over 27 held-out curves. This is progress against the incumbent, not a claim that the weighted stage and the complete variant were one already-measured stack.[ROUND1][ROUND2]

7.3 Degree five: a large stage factor with a distant model crossing

The quintic Joux–Vitse/torsion programme implements a four-point decomposition test over $\mathbb{F}_{p^5}$. Its plain and Edwards-symmetry systems have very different costs: the cited constants are roughly $3.0 \cdot 10^{10}$ and $7.3 \cdot 10^7$ base-field multiplications, a recorded stage factor around 416. But a corrected exact yield census changes the residual rate from $1/(24p)$ to $1/(192p)$, worsening the extrapolated crossover. A four-torsion follow-up then retracts an anticipated degree-halving lever: the available translation does not implement the needed involution over the stated base field, and the measured saturation costs about as much as it gains in rate. [QUINTIC]

These examples explain why a several-hundred-fold algebraic improvement can coexist with an impractical complete-method prediction. The programme has learned both a useful symmetry implementation and the conditions under which its hoped-for scaling argument fails. Neither observation should be compressed into simply “torsion works” or “torsion does not work”.

7.4 Covers and genus-three index calculus

The cover programme implements extension fields, Cantor-style divisor arithmetic, a GHS/Joux–Vitse-type cover, a point transfer and a decomposition test checked against a separate enumeration oracle. On the favourable weak class, reducing the oracle constant improves the complete IC/rho ratio as size grows. It reproduces a known attack route; the difficult additional question is whether a generic source can find

and reach such a class at a total cost below its own rho reference. [COVER]

The more recent cubic-GHS/genus-three work adds double large primes and new matrix strategies. The first improvement round reports a complete cost exponent about 1.385 rather than 1.910 on a common $q = 2^{10}$–2^{18} range. Earlier headlines compared different ranges and are corrected. The candidate's local exponent rises at larger sizes, so a global finite-range fit is not evidence that it has reached the asymptotic 4/3 prediction.[ROUND1]

A separately rebuilt genus-three baseline reports IC/rho around 0.252 at $q = 2^{14}$ after the study's rho normalisation. That comparator is rho on the same Jacobian subgroup, not rho on the original elliptic curve before transport. The timeline explicitly labels it as a Jacobian result. An ECDLP crossover requires the source transfer and source reference to be in the comparison.[TIMELINE]

The later structured-elimination and block-Wiedemann rounds reduce complete costs further, but also identify the relation phase and matrix growth that remain. This is the strongest recorded fitted-exponent improvement over an incumbent among the selected complete toy pipelines. It is appropriately described as reproducing and engineering a known higher-genus route at toy scale, not discovering a new exponent for general prime-field ECDLP.[ROUND2]

8 Isogenies, graphs, fibres, conductors, and transfer

The isogeny work is not a single failed shortcut. It contains several different objects: explicit walks and map certificates, conductor and endomorphism-ring calculations, fibre-aware bases, gluing and higher-genus constructions, weak-node searches, graph-wide experiments, and cryptographic scheme prototypes. Their obligations differ.

8.1 What transport conserves

For a homomorphism $\varphi : E(\mathbb{F}_q) \to E'(\mathbb{F}_q)$, let $\kappa = \ker \varphi \cap E(\mathbb{F}_q)$. Suppose $\mathcal{B}' \subset \varphi(E(\mathbb{F}_q))$ and take the complete rational pullback $\mathcal{B} = \varphi^{-1}(\mathcal{B}') \cap E(\mathbb{F}_q)$. The recorded transport argument states that the number of ordered m-decompositions of R over $\mathcal{B}$ is $|\kappa|^{m-1}$ times the number of decompositions of $\varphi(R)$ over $\mathcal{B}'$. Each image tuple has $|\kappa|^m$ lifts, and exactly $|\kappa|^{m-1}$ have the required sum.[TRANSFER]

This is a useful scoped statement. It applies to the declared homomorphism and complete pullback domain. It does not prove that every alternative factor-base geometry has the same cost, nor that every non-homomorphic or higher-dimensional reduction is impossible. It directs attention to three things that can still change: rational kernel symmetry, destination geometry, and arithmetic/solver cost. Map evaluation and source recovery remain additional costs.

Isogeny loops act as endomorphisms; on a fixed prime cyclic subgroup an applicable endomorphism acts by a scalar. Faster evaluation can help known-scalar multiplication or a generic walk. Turning it into a non-generic unknown-log advantage requires another mechanism. The portfolio correctly keeps these claims distinct in its newer transfer and curve-identity rules.[FIBRES][C-RULES]

8.2 Conductor gaps and fibre-aware IC

The conductor-gap programme examines whether endomorphism-ring levels change the measured difficulty of a decomposition method. Two fixed CM pairs with large prime gaps give recorded floor/crater counted-S ratios 0.975 with 99% interval [0.603,1.451] and 1.127 with interval [0.922,1.342], across 192 verified records and 96 stored exact replays. These observations do not resolve the general conductor-gap question; they fail to detect an effect in those exact comparisons. Both sides' unpriced work remains relevant.[CONDUCTOR]

Fibre closure and two-torsion symmetrisation can reduce redundant algebra on a matched abscissa base. A later correction shows why the domain must be specified: at matched column count, an apparent symmetry saving can disappear. Similarly, the fibre-aware scaling panels improve a specific method but show its IC/rho envelope widening with size at fixed cofactor. The reported ratios rise roughly from 8–9 to 35–37 at cofactor four, and from 13–20 to 60–63 at cofactor 72. [FIBRE-SCALE][PRIOR]

The right question is therefore a controlled comparison on the same transported target, base convention, subgroup and resource envelope. Comparing two curves with different subgroup sizes or comparing an orbit-redundant base to an already-quotiented base does not isolate intrinsic hardness.

8.3 The graph-building capability and its measured coverage

The full-volcano work builds explicit finite graphs and routes, exports explorers, verifies kernel maps, assigns conductor strata, and runs bounded IC and generic solvers on selected vertices. This is a substantial implemented capability. Graph generation, graph completeness for a declared edge set, and graph-wide attack execution are three different completion claims.[VOLCANO]

The accessible checkpoint covers three smaller classes with 24,823 declared vertices and 51 strata. The repaired checkpoint has 2,935 audited vertices: 989 in the mixed-conductor graph, 970 in the three-power graph, and 976 in the two-power graph. All 51 strata have samples. This is about 11.8% of the declared vertex population, not a completed graph-wide measurement.

Table 4: Outcomes in the downloaded repaired checkpoint. They are attempts across multiple arms and two target workloads per selected vertex, not 46,960 independent curves. Source: [VOL-ARCHIVE].

Quantity	Recorded count
Audited vertices / declared graph vertices	2,935 / 24,823
Strata sampled	51 / 51
Attempt records	46,960
Verified recoveries	30,270
Exhausted budgets	14,626
Timeouts	2,064
Relations found	1,530,557
Source relation trials	74,829,850
Actual decomposition calls	71,296,000

Source trials and oracle calls differ because repeated queries can be skipped before invoking the oracle. Timeout counters are unknown where not emitted; they are not zero. Earlier report versions labelled source trials as decompositions, and the corrected summary exposes both denominators. Those corrections are a practical example of why raw records and versioned summaries matter.

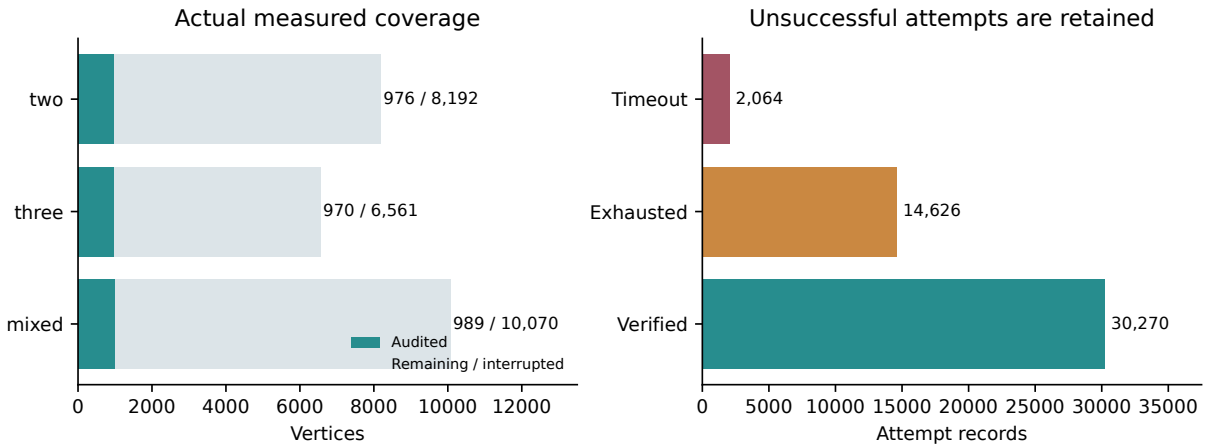


Figure 5: Coverage and outcomes of the actual 2,935-vertex checkpoint. The left panel compares audited with remaining vertices in each of the three finite graphs; the right panel preserves unsuccessful attempts. This is a coverage chart, not an inference that unsampled vertices share the same outcomes. All 51 strata are sampled. Source: [VOL-ARCHIVE].

The three source models have very different largest prime subgroups: 61,751; 19,768,889,568,671; and 4,253. Their roughly 57–59-bit field sizes must not be used as common subgroup security sizes. The original A–E ambitions are much larger—about 7.83 billion expected vertices and 515 strata—and remain separate incomplete work. The smaller checkpoint cannot be described as completing those populations.

This review downloaded the archive and checked every one of its 29,409 manifest entries, with no missing entry or hash mismatch. That proves byte consistency of this delivery. The checkpoint's solver audits are same-implementation replays, not new independent validation by this review. Its unqualified wall times and incomplete operation counts do not establish a fastest-method or asymptotic claim.

8.4 GHS mechanisms and the scope of a trapdoor demonstration

The pinned main branch of `ghs-trapdoor` is a scaffold, but another captured branch contains a Sage mechanism prototype. It describes a composite-degree binary field $\mathbb{F}_{2^{20}}$ over $\mathbb{F}_{2^5}$, a weak curve, an Artin–Schreier tower, a genus-four model, an explicit transfer, and a small protected subgroup of order 7,717. The branch reports homomorphism and subgroup-order checks, isogeny pullback, and end-to-end toy ElGamal recovery.[GHS-MAIN][GHS-BRANCH]

Those are mechanism claims on the branch, not a demonstrated scalable escrow system. The toy destination solver is BSGS, whose square-root cost is comparable to source rho at that tiny subgroup order. The general claim that the public curve forces outsiders to use rho, or that a secret chain is hard to replace with any useful public route, requires additional analysis. Working-field ambiguity in the README's Jacobian description also needs checking against the actual code and certificates before a stronger transport claim is made. No fresh Sage or Lean replay was performed here.

The distinction is especially important for ECC2K-130, whose extension degree 131 is prime. A construction exploiting a degree-four extension cannot simply be scaled by changing that parameter. The ten-lead register contains finite order and map-class obstructions for particular candidate Jacobians, but leaves other destinations and constructions open.[TEN]

8.5 What remains of the structural and hidden-route leads

The structural-completeness draft organises seven families of known isogeny/CM mechanisms. Its own correction now describes case-analysis proof sketches that have not received independent review; an earlier unconditional-proof sentence was removed. This paper therefore treats it as an organised argument and source catalogue, not a proof that every prime-field attack route has been exhausted.[STRUCTURAL]

A hypothetical hidden route should be assessed through separate questions: mathematical existence, explicit construction, public discoverability, subgroup action, destination advantage, evaluation and recovery costs, and the availability of an equivalent public shortcut. No inference about secret agency capability follows merely from a large degree or a sparse weak class. Conversely, failing to make a route practical in a toy study is not a proof of its universal impossibility. These are the appropriate remaining obligations of the programme's speculative endomorphism and isogeny leads.

9 Learning, symbolic discovery, and experimental falsification

The ML repository is more than an attempt to predict secret scalars. It implements curve and isogeny-class datasets, exact arithmetic features, class-held-out learning, graph models, symbolic rule search, permutation nulls, and certificate-producing follow-ups. Its useful outputs include both rediscovery of planted structure and evidence that attractive hardness correlations disappear under stronger controls.[ML]

9.1 The hardness question after controlling for size

The expanded M9 hardness report has 108,473 nodes in 377 isogeny classes, over field sizes mainly 16–24 bits. It distinguishes the counted cost of the C library's rho, an exact two-sum yield label, planted structural labels, and within-class residual versions of those quantities. Whole isogeny classes, field families and larger sizes can be held out. [ML-HARD]

A large global R^2 is not evidence of a new hardness predictor. For plain-rho cost on the isogeny-class split, the full model's R^2 is 0.884 while the ceiling-only baseline is 0.885. For negation-rho it is 0.864 against 0.863, with a skill interval including zero. Much of the global predictability is the known size and automorphism law. On within-class blind cost labels, the added skill is essentially zero or negative in the reported panels.

The positive controls do work. The within-class planted-rho label has $R^2 = 0.914$ for the planted-safe model against 0.918 for its oracle; other planted folds are recovered strongly. Thus the experiment can detect a deliberately supplied structural law. It does not find a corresponding blind law that improves the measured IC pipeline.

The report also exposes power, split design and feature leakage. A model using walk-sampling features can learn which node was planted as a root rather than a transferable mathematical property. A learned

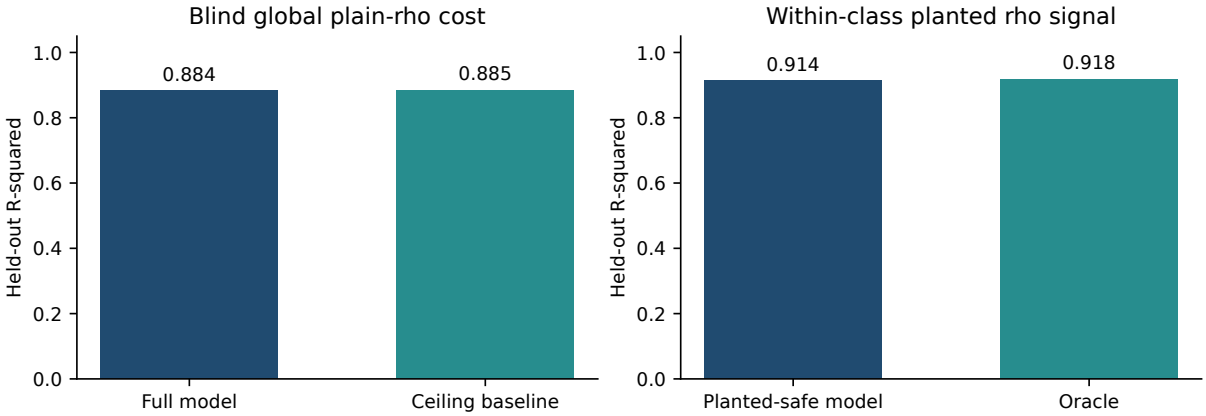


Figure 6: Why a high prediction score alone is insufficient. Left: global plain-rho cost is already predicted by the simple ceiling baseline. Right: a within-class planted structural control is recovered near its oracle. These are different labels and are not a comparison of two attack speeds. Recorded M9 values on the isogeny-class split; source: [ML-HARD].

policy must eventually be charged for features, inference, selection and missed solutions inside a held-out complete pipeline. No such learned component is shown here to improve the current measured IC frontier.

9.2 Binary GHS labels vary, but they are formula labels

The GHS study covers 507 classes and 114,453 nodes over fourteen fields. It computes exact structural invariants and applies a declared literature cost formula. The descent label varies within isogeny classes through the curve coefficient; planted weak nodes and subfield-type structure are not the same phenomenon.[ML-GHS]

On held-out classes, coefficient invariants identify the weak-node label much better than intrinsic graph features: reported in-class AUCs are about 0.98 and 0.60. The subfield-degree/endomorphism-level relation has zero violations on 99,269 checked nodes in the report. These are useful structural observations and tests of a formula model. They are not measured complete GHS attack times, and they do not establish a secret-to-public discovery asymmetry at cryptographic scale.

9.3 Extension-field degree measurements

The $\mathbb{F}_{p^n}$ study covers 453 curves over degree-two and degree-three extensions, with random, subfield, twist, isomorphic and isogenous families. Its 5,592 target/system rows have Macaulay ranks equal to the reported regular-sequence values, with the stated solving and refutation-degree behaviour. It finds target-level degeneracy and model-over-subfield effects, rather than a general lower degree from sharing an isogeny class.[ML-FPN]

This is evidence about particular formulations over the tested primes. It should not be pooled with Boolean first-fall experiments as if they measured the same ideal or the same degree. It also explains why "isogenous" and "same decomposition geometry" must be distinct features in a predictive experiment.

9.4 Symbolic discovery with exact follow-up

The Smith genus-three programme constructs and studies rational $(2, 2, 2)$-isogeny mechanisms on 140,000 random squarefree octics. Its observed density of curves admitting the declared rational isogeny is 0.1864, with interval [0.1843,0.1884], near the cited prediction 0.1857. It separately tests trigonal-map rationality, isogeny rationality and independence across pairings.[ML-SMITH]

The rule miner searches products of explicit quadratic-character features and evaluates candidates on held-out primes and label permutations. The follow-up twist-bit report has 5,811 certified nondegenerate cases with no counterexample to the stated character law; 266 trace-zero cases are explicitly not certified by that point-count criterion. Algebraic identities are checked on additional large-prime panels. This is much stronger than reporting a classifier score, because it produces a readable candidate rule and a

certificate boundary. Finite validation remains distinct from a proof for all permitted curves.[ML-TWIST]

The Humbert-surface work recovers equations for $H_4, H_5, H_8, H_{12}, H_{13}$ from structured source points, checks them against published equations, and tests fresh primes. The recorded least weights are 30, 24, 60, 120 and 120 in the chosen invariant presentation. This is a successful rediscovery and validation workflow, not a priority claim for new Humbert equations. It also corrects an overgeneralisation: rational real multiplication in a particular family cannot be inferred for every point of a Humbert surface from the residue symbol (D/p) alone. [ML-HUMBERT]

9.5 Assessment of the learning leads

The most useful learning output so far is an instrument for generating explicit, falsifiable structural candidates and then removing leakage and simple explanations. The main hardness-prediction result is negative beyond the known baselines at the measured scale; the symbolic and planted-control results are positive within their declared tasks. Future learning should target a concrete complete-method decision, such as a base or solver policy whose feature cost is charged, rather than a large global R^2 with no effect on ECDLP work.

10 What was built around the mathematics

10.1 A portfolio of executable instruments

The Rust library implements field and curve arithmetic, generic and specialised discrete-log methods, factor bases, Semaev oracles, SAT and matrix engines, transfer experiments, external-CAS exporters, cost accounting, curve identity and visual tools. The native C library provides 64-bit reference groups and solvers, statistics, batched curve arithmetic, preprocessing, bindings and distributed interfaces. These 64-bit primitives are research instruments; their parameter limit is not a claim of cryptographic-size support.[C-README][CA-README]

The broader library also contains symmetric ciphers, hash functions, signature and nonce demonstrations, RSA and factoring routines, pairings, zero-knowledge examples and post-quantum educational implementations. Their presence documents capability breadth, not an audited production cryptographic stack. For example, the current README explicitly labels the STARK-like modules as educational, unaudited and unsuitable for supporting a research security claim. This review does not turn a module inventory into a security certification.

WDSat and the SQIsign repository are upstream solver/protocol forks and integration resources; their imported histories are not independent algorithmic contributions by this programme. The small standalone ecbench repository is a companion entry point, while the principal measuring implementation lives in crypto. Isogeny-labs supplies a presentation layer. FPGA work includes field, point, step-unit and walker models and simulation targets; the code inventory alone establishes no measured silicon throughput. These distinctions prevent repository proliferation from being counted as an equivalent number of new solvers. [C-README][CA-README]

The generic-group cost challenge adds a particularly useful boundary: the solver receives opaque point tokens, while an oracle owns the real coordinates and operation counter. This prevents an implementation from silently doing unmetered point arithmetic. The same discipline motivates the later explicit counters and sealed session records, although a coordinate-aware IC study is a different computational model. [COST]

10.2 Research automation and preservation

The autoresearcher separates coordinator, idea generator, executor, reviewer, validator and red-team responsibilities. It implements protocols, role policies, handoffs, immutable records, archive receipts, corrections, claims, knowledge retrieval, runbooks and resource coordination. This is research infrastructure whose aim is to prevent an observation from being mistaken for an official scientific transition.[AR-README]

The freshly parsed direct `ledger/goals/` files contain 72 goal records: 36 draft, 26 active, six completed and four closed at budget. These are literal stored status fields, not this paper's judgement of scientific completion. Direct hypothesis files include 2,124 parsed records; only 32 have exactly the literal status

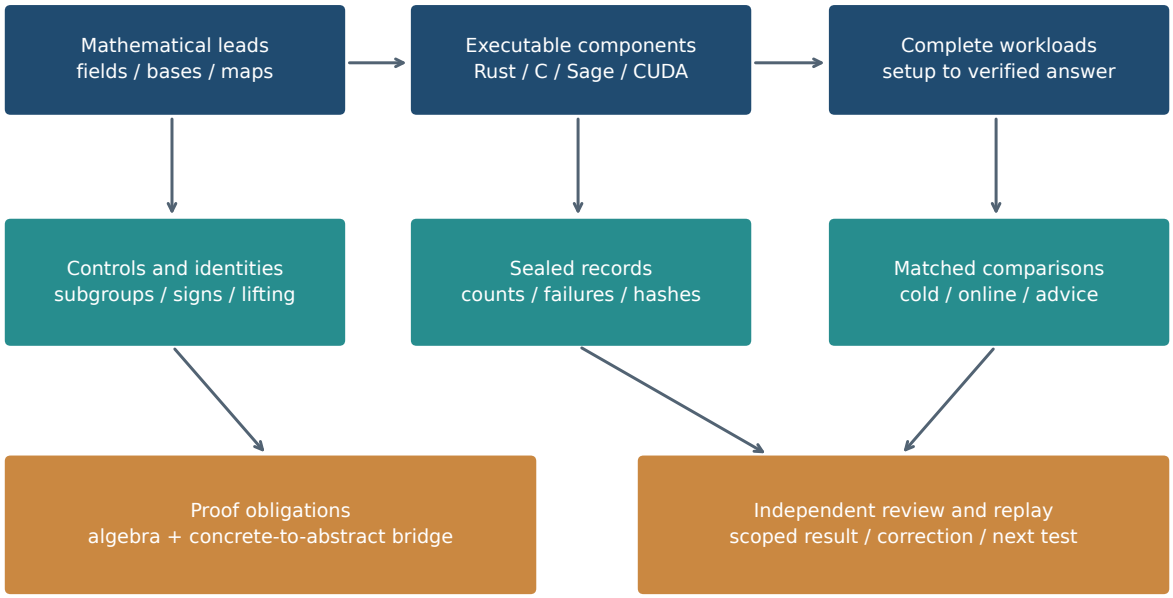


Figure 7: The implemented research stack and the distinct obligations between a mathematical lead and an accepted complete-method claim. Arrows are a conceptual workflow, not an assertion that every historical study passed every gate. The catalogue and source register link the concrete implementations behind each box.

`supported`. Other scoped and custom statuses are retained in the catalogue rather than forced into a binary success metric. Legacy root records account for why a whole-ledger wrapper count gives different totals.

The direct directories also contain 6,256 parsed handoffs, 2,634 proposals and 187 questions; missing status fields are common. These counts measure coordination volume. Without deduplicated chains from proposal to execution to independent review, they cannot yield a meaningful "research success rate". The 24 safe-loader failures and explicit correction records identify maintenance work rather than silently disappearing from the denominator.

Cairn and associated runtime tools provide execution and custody coordination. Their source and commits belong in an implementation assessment, but this review does not infer actual rented compute, financial spend, agent-hours or effective parallelism from their existence. No verified consolidated bill or complete compute ledger was available for that purpose.

10.3 GPU rho, CPU arithmetic, and challenge operations

The Blackwell release records an audited campaign build at 14.64 billion rho iterations per second on an RTX PRO 6000 Blackwell, with a later best build at 15.88 billion. Its central mechanism is accelerated binary-field arithmetic using the carry-less multiplication exposed by the cited CUDA/PTX toolchain. Those are historical hardware-throughput claims with provenance to the three central repositories; this review does not remeasure them.[RHO-GPU]

The README also distinguishes the developed measurements from the release snapshot's validation: the CPU client and planted solves were run from that copy, while its Docker image and GPU modes were not yet run there. The multi-card completion times are estimates from a work model, not completed ECC2K-130 recoveries. Different documents quote roughly 4.3 or five card-years under different builds and constants; they must retain those assumptions rather than become a single measured campaign duration.

The `ecc-challenges` service adds an operational evidence chain: versioned raw uploads, namespaces, admission receipts, record validation, point-relation checks, indexing and collision recovery. A durable upload ACK is not the same as verified work. The v2 path checks curve membership and $R = [a]G + [b]Q$ before index insertion, while reported trail lengths still lack a work-verification model. Consequently point

counts and upload volume cannot honestly become percentage completion of a challenge.[ECC-SERVICE]

An open C-library branch also identifies fruitless cycles in the GLV walk and reports a corrective implementation. Its paired operation ratios are 0.658 for $j = 1728$ and 0.768 for $j = 0$, with Bonferroni-adjusted 98.75% intervals [0.630,0.688] and [0.753,0.784]. The author reports 432,000 correct toy solves across three builds. These are unmerged, operation-count observations on 24–32-bit subgroups; the change does not prove a new ECDLP hardness law. A stronger generic implementation also changes the reference against which future IC must be measured. [PR953]

10.4 Quantum circuits and scheme construction

The quantum point-addition work log studies reversible arithmetic scored by average Toffoli count times peak qubits. Its recorded baseline is 1,739,149 average Toffolis, 1,434 qubits and score 2,493,939,666 over 9,024 validation shots. It also records truncation and reroll screening. Those observations concern simulated circuit resources and finite validation. Adaptive screening for a favourable Fiat–Shamir test set does not prove correctness on every input. Upstream challenge progress must not be credited as this programme's independent circuit advance, and no fault-tolerant hardware ECDLP execution is demonstrated. [QUANTUM]

EigenX is a separate constructive output: a Rust KEM/PKE research implementation, wire formats, parameter sets, known-answer and hostile input tests, and a Kani-chain/endomorphism design. Its current README separates classical-optimised and quantum-hardened claim tiers after a correction to earlier category language. Correct decapsulation at the nominal parameter sizes establishes functionality, not the conjectured security level. This is a scheme-construction research line and is not evidence of an ordinary ECDLP break.[EIGEN]

Finally, `cryptologic-archives` preserves international standards and historical cryptologic sources with claim and source catalogues. It contributes provenance and contextual research, while its documentary counts are not algebraic ECDLP experiments.[ARCHIVES]

11 What is proved, observed, conditional, or still unproved

11.1 A proof map with explicit bridges

The portfolio has moved from isolated Lean files toward a shared `CryptoLean` library and a repository-wide proof gate. The shared library pins Lean 4.35.0 and a Mathlib revision; historical dependency-free projects can use other pins, such as the scoped torsion workspace's 4.22.0. A blanket statement that every proof uses one verified toolchain would therefore be incorrect.[LEAN][TORSION]

The gate builds targets and audits transitive axioms, accepting standard foundations and rejecting `sorryAx`, unlisted native evaluation and user axioms. Its README correctly states that passing this gate does not establish that the formal theorem is the report's theorem. A faithful mathematical bridge and an independent statement review remain needed. This review inspected central proof sources and receipts, but did not run a fresh Lean kernel build; it does not issue a new kernel certificate.

Result or argument	Supported statement	Remaining boundary
Semaev S_3 roots	The committed Lean source proves the displayed affine algebraic factorisation and root equivalence under curve equations and distinct abscissae.	Coordinate/chord definitions are visible; exceptional points and the relevant curve-model interpretation still belong to the implementation contract. No fresh kernel build here. [RF-PROOF]
Odd-torsion translation	Under the stated additive laws, odd r, $[r]P = 0$ and nonzero two-torsion T imply $[r](P+T) = T \neq 0$.	The new generic proof is distinct from the imported curve application. No concrete point model or factor-base membership checker is imported by that file. [TORSION]

Result or argument	Supported statement	Remaining boundary
Transport conservation	A homomorphism with a complete rational-pullback base has the stated kernel-fibre decomposition count.	It is a scoped written argument, not a new audited Lean theorem here or an exclusion of all different bases and reductions. [TRANSFER]
Coprime group orders	A homomorphism between finite groups of coprime orders has trivial image; the register applies a finite order check to named source/destination pairs.	It excludes those homomorphism routes, not every higher-genus destination or every reduction. [TEN]
Smith/twist and Humbert laws	Explicit identities, published-equation comparisons and finite certificates support the stated populations.	Zero finite counterexamples and a checked interpolation do not alone prove every global mathematical extension. [ML-TWIST][ML-HUMBERT]
Structural completeness	A seven-family taxonomy and case-analysis proof sketches with an explicit October correction.	Independent review and stronger proofs remain due; no universal security theorem is established. [STRUCTURAL]
Degree plateaus and exponent fits	Exact outcomes on frozen bounded systems; fitted exponents on named size ranges.	No proved global degree law or cryptographic-scale asymptotic improvement. [DREG][FRONTIER]

The shared library itself lists unfinished bridges, including the relation between its concrete binary-polynomial computation and the Mathlib polynomial object, group-level isogeny and generic-group definitions, and additional Semaev/degree material. These omissions are explicit research obligations, not proof failures to hide.[LEAN]

11.2 Corrections that change the scientific reading

1. **A solver label was wrong.** The several-thousand-fold NIST query ratios are against Buchberger, not the claimed F4 comparator. The large effect remains a stage observation with the corrected label. [RF-AUDIT]
2. **A complete-solve label hid composition.** The larger prime msolve totals were composed estimates from sampled per-query cost, unlike the directly timed 16-bit comparison.[RF-AUDIT]
3. **Unpriced algebra changed the ranking.** F4 XORs and F6 geometry can dominate cost after being included. A lower algebraic counter is not a lower total.[GB][C-RULES]
4. **Online preparation changed the comparator.** The historical IC online ratios remain observations, but matched preprocessing rho removes the claimed advantage under the later controls. The exact wide equal-bytes protocol is still distinct and unrun.[EQUAL][EQUAL83][WIDE]
5. **Relation count was not rank.** The free-collision census finds far fewer independent rows than collision counts suggest. A later tuned production improvement survives on a smaller constant boundary. [EQUAL][ROUND2]
6. **A trace-zero control sampled the wrong population.** The apparent 50% rejection does not apply to the production odd-order IC targets. The integrated filter makes zero rejections in 29,624 checks and contributes no such gain.[ROUND1]
7. **LLL failures were misattributed.** The corrected note withdraws curve-specificity and identifies floating-point overflow for the relevant cases. It also withdraws an unsupported P-521 mechanism: the fixed-point recovery stands, but the scaled-floating failure's exact cause remains unresolved. Repeated deterministic probes are not distinct instances.[LLL]
8. **The structural proof was overstated.** The manuscript now admits proof sketches rather than an independently reviewed unconditional theorem.[STRUCTURAL]
9. **An Semaev expression defect may affect historical results.** An open matched legacy/repaired rerun is designed around the archived call sites. Some trial packages exist; the final validation and ledger transition are not complete in the observed PR snapshot. Older exposed results remain integrity-qualified until that audit resolves them. [PR2412]
10. **A simulated campaign is not a measured one.** The open synthetic-volcano PR explicitly labels its 130,410 observations as invented cost-model records with no actual curves or recovered logs. It is useful

planning and presentation tooling and is excluded from this paper's empirical attempt counts.[PR2236]

These corrections should remain searchable beside their original claims. Suppressing superseded evidence would make future reviewers repeat the same misinterpretations. Conversely, a correction does not erase a separately valid measured improvement whose boundary was never the corrected one.

11.3 What the combined record does not prove

No inspected evidence establishes a general classical prime-field ECDLP algorithm with exponent below one half, a cryptographic-scale IC solution of ECC2K-130, or a method that recovers a standard deployed curve's arbitrary unknown logarithm from the public point alone. The record also does not prove the security of those curves or the impossibility of a future non-generic attack. Generic-model bounds, finite search failures, component improvements and implementation limits have different logical force and must not be substituted for one another.

12 Lead assessment and the next discriminating work

12.1 A concrete progress ledger

The table below collects examples that answer "what improved?" without collapsing unlike tasks into a single progress score. Ratios are candidate/incumbent unless explicitly stated otherwise. A value below one means less of the stated cost, not necessarily a faster complete ECDLP. No cumulative factor is obtained by multiplying rows with different inputs, baselines or boundaries.

Concrete change	Recorded progress	Boundary and evidence	Remaining limit
Production free relations	0.960 [0.946,0.976]; 360 verified runs	Tuned complete counted IC comparison; engineering. [ROUND2]	Small gain; not a sub-rho result
N83 smaller compact base	2.645 s saved [2.085,3.205]	Five-point cold-process holdout; engineering. [N83]	Two leading candidates not significantly separated
Weighted D4 solver	About 18 times less than previous invariant form	Decomposition constant, several primes, root checks; engineering. [D4]	No complete-method S in that study
D4 first-dependency stop	0.979 [0.973,0.986]	Complete counted comparison on 27 held-out curves. [ROUND2]	Extension-field toy regime
Genus-three matrix route	0.653 [0.647,0.660] at 2^{20}	Complete counted solve; 474 logs verified. [ROUND2]	Phase exponents and source-to-Jacobian comparison remain distinct
F4 large matrix on RTX 5090	425.8 s to 5.70 s	Same task, ranks, refutations and pinned variables. [F4-GPU]	Stage only; inherited CPU wins another oracle boundary
H100 tiled update	1.11–1.49 times faster	Historical device-stage measurement; engineering. [F4-PANEL]	Upload and full-query costs remain
Inherited F4	198.6–222.1 times less matrix work	Identical-tree stage comparison. [GB]	Not a uniform general-CAS speed claim
Factor-base degree predictor	MAE 0.06; 0.08 leaving out fields	Empirical rule on 656 bases. [HEUR]	Not a proved scaling law

Concrete change	Recorded progress	Boundary and evidence	Remaining limit
Symbolic twist-bit rule	5,811 certified cases, zero counterexamples	Finite certificate-backed observation. [ML-TWIST]	Global theorem and independent semantic review are separate
Graph-wide instrument	2,935 audited vertices, 46,960 attempts	Actual checkpoint; manifest freshly checked. [VOL-ARCHIVE]	11.8% of declared smaller graph population
Native-toolkit reproducibility	17 configured tests covered	Fresh Release build, 15 initial passes plus two repaired-environment passes	Core native tests, not every historical experiment

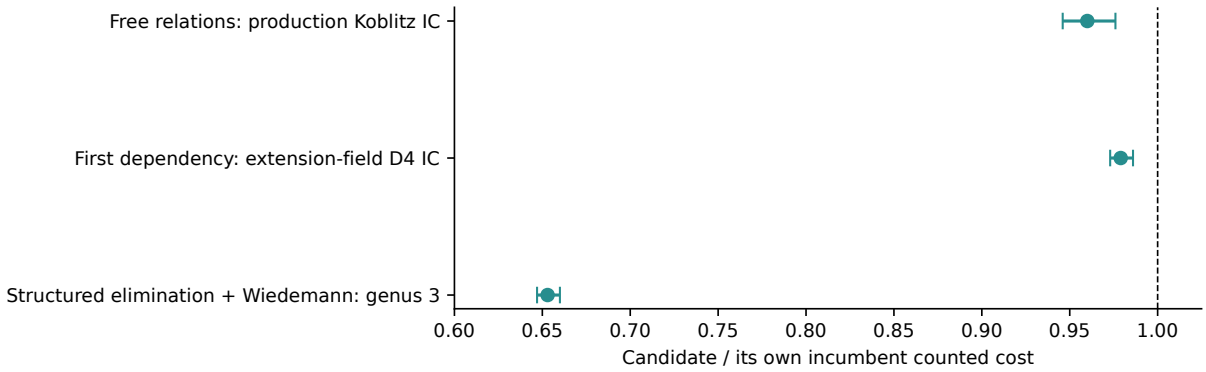


Figure 8: Selected recorded candidate/incumbent comparisons. Each row keeps its own task and baseline: the chart is a compact ledger, not a ranking of methods or a cumulative speedup. Intervals are those reported by the underlying studies. Sources: [ROUND2].

These examples support four concrete conclusions. The programme has built working instruments rather than only speculative notes. It has made verified improvements over its own IC incumbents. It has developed stronger controls that reverse some attractive interpretations. And it still has a measured gap between its complete IC methods and the strongest matched generic reference in the principal cold campaign.

12.2 The ten open leads, without blanket closure

The public ten-lead register preserves nineteen advisory records. Every lead remains open in that register and no official hypothesis status is changed by this paper. The observations below are what the register reports; its private supporting packets were not freshly replayed here. The proposed next steps are discriminating tests, not scheduled runs or completed experiments.[TEN]

Lead	What the record supports	Next discriminating obligation
1. Higher-arity refutation	A full-width timeout and memory error were followed by compact row and kernel methods deriving one at degree four on a specific negative system. Other degree-three closures do not refute their named fixtures.	Freeze new negative and satisfiable controls; compare full row spaces, proof sizes and complete costs. Do not infer a minimum refutation degree from one closure.

Lead	What the record supports	Next discriminating obligation
2. Frobenius-equivariant F4	The register's ordinary F4 fixture is checked, but its complete character-block solver is not implemented. Separate G2 passes elsewhere do not discharge that construction.	Build one reconstructible character-block elimination stage over the declared splitting field and verify the entire row space against F4.
3. Prime five-sum common factor	Six endpoint/evaluation arms agree on the frozen queries; cached construction reduces counted calls in the reported panel. The producer still enumerates B^3 endpoint occurrences.	Seek an explicit construction avoiding that enumeration; test larger primes with cold and warm setup charged. Tiny-field saturation is not a scaling argument.
4. Reachable weak curves through covers	Eight declared rational two-isogeny components over $\mathbb{F}_{7^6}$ are exhausted with no hit for one norm-one branch-ratio criterion. A twist-commutation identity is scoped to its stated transition.	Implement a concrete new transition, independently verify its map and subgroup action, then search with smoothness and field-of-definition gates.
5. Small-norm endomorphism closures	Two matched-advice methods recover zero targets in 32 held-out attempts within 512 online steps; advice logs are checked in the source record.	Require a structural coverage prediction before enlarging the bank; compare norm and walk banks with matched memory and charged setup.
6. Higher-arity torsion symmetry	The subgroup-only domain is not stable under nonzero two-torsion translation at odd r. Suitable ambient domains can be stable.	Construct a T-closed mixed-domain base and quotient equations; compare to the same unquotiented domain and verify every lifted solution.
7. Degree-131 higher-genus destination	The declared census produces 22 smooth models and five Weil classes. Finite order checks obstruct particular homomorphisms for source primes above 263.	Produce an explicit nonzero source/destination map with the required subgroup before another candidate census; other targets remain open.
8. GPU F4 route	Seventy-six matrix/thread configurations pass host device-source emulation and row-space replay in the register. Actual GPU evidence elsewhere belongs to different tasks.	Run this frozen panel on a device, then a complete-query paired comparison including transfers, setup and replay.
9. Degree-83 relation control	Rank 600 and eight target logs in a 53-bit subgroup are reported using 32 imported and 568 new rows.	Rebuild an entire smaller rank instance from zero and charge relation, rank and target costs against rho. Imported work is not free.
10. XOR/PB curve-lift filtering	The register preserves filtered resolutions and thirteen target precheck rejections, separately attributed; satisfying an encoding alone is insufficient.	Freeze a matched target panel, randomise arm order, price filter construction, verify point lifting, and repeat under qualified isolation.

A timeout leaves an implementation or resource question; a finite enumeration excludes its declared finite domain; a group-order argument excludes its stated map class. Reopening a scoped obstruction requires a changed load-bearing assumption or a challenge to the proof, rather than merely a new name for the same construction. Conversely, no one of these outcomes closes an entire mathematical direction.

12.3 Priorities justified by the evidence

First: complete the accounting and integrity bridges. Resolve the legacy S4 defect-impact audit, maintain root-call and native counter emission on every arm, and co-measure the exact wide equal-bytes preprocessing control. These actions determine what existing figures mean; they do not require promising a new exponent. The audit must retain environment drift and legacy-reproduction failures rather than assigning every difference to the repaired expression. [PR2412][WIDE]

Second: answer one genuinely discriminating degree question. The $m = 4, \ell = 6$ closure cell separates the two surviving toy explanations. The PKM $m = 3, N = 21$ trace now shows why a somewhat larger host alone is not automatically enough: the next dense object has a derived 317-GB payload before overhead. A new sparse or blocked engine must reproduce the same closure and controls; a larger host must retain the same protocol and an explicit stop. Neither approach should change the mathematical question simply to make a run finish.[DREG][PR2160]

Third: integrate component improvements into complete IC. The factor-base product-profile rule, root harvesting, calibrated sparse linear algebra, weighted invariant systems and selected GPU stages are concrete candidates. Each needs a held-out comparison to the current incumbent with every phase charged. A positive component result can be reported immediately at that boundary; it becomes a method improvement only after the complete comparison moves.

Fourth: pursue transfers through explicit maps. Graph coverage can expand, but the next decisive transfer experiment should name a source subgroup, a verified route, a destination solver and a source-to-answer comparator. A larger collection of low-genus objects without a subgroup bridge is not a substitute. For a graph effect, hold subgroup size, target construction and base convention fixed or model them explicitly, and keep sampled coverage separate from complete populations.

Fifth: make learning choose a paid decision. The positive symbolic results justify using ML as a candidate generator. The blind hardness nulls argue for targeting a specific useful decision rather than another unadjusted global score. A base or solver policy should outperform a simple constant policy on held-out classes after feature extraction, inference and failures are charged. Any learned rule that becomes a mathematical claim needs an exact statement and independent verification beyond predictive accuracy.

Sixth: strengthen the proof-to-report map. Build the concrete-to-abstract bridges listed by CryptoLean, inspect transitive axioms, and independently check that each report theorem has the same quantifiers, exceptional cases, fields and subgroup hypotheses as its formal declaration. Formalising a smaller algebraic identity is valuable; it must remain labelled as that identity if the full claimed cryptographic implication is still open.[LEAN]

12.4 What the programme has learned

The central lesson is operationally precise: the bottleneck can move, and the comparison must move with it. Eliminating expensive polynomial machinery can expose relation yield; faster relations can expose rank generation; faster rank generation can expose setup or target work; and a better generic reference can erase an apparent method advantage. The programme's strongest progress is the combination of implemented improvements and instruments able to show that a preferred explanation was wrong.

The result is a broad collection of verified toy mechanisms, useful algebraic and systems components, measured incumbent improvements, structural conjectures with finite evidence, and named open obligations. It is a foundation for continued ECDLP research whose next successes can be measured against explicit baselines. It is not yet an established general ECDLP exponent advance.

A Reproduction, coverage, and deliverable status

A.1 Companion artifacts

The paper directory contains editable LaTeX, the built PDF, vector figures with PNG browsing copies, their recorded input data, a pinned source register, compact review receipts and a compressed catalogue. The catalogue contains repository and branch manifests, commit metadata, tracked-file inventory, document and study indexes, literal ledger status fields, parsing failures and topic-tag definitions. The tags are overlapping textual classifications; they are not scientific judgements or independent experiment labels.

The separate downloaded volcano archive remains linked through its original Drive identity and SHA-256. The paper does not copy its executables or raw solver tree into a new experiment, and no downloaded binary was executed for the archive check. The byte-manifest receipt states exactly which entries were verified.

A.2 Fresh checks and their limits

1. Git trees and snapshot-reachable histories are enumerated at the recorded pins. All current captured heads are listed separately, with their own metadata union. A post-snapshot open branch never becomes landed evidence merely by being fetched.
2. All nineteen frontier `records.jsonl` hashes match their session metadata. Fresh counting reproduces 4,848 total and 3,232 non-warm-up executions, including their unsuccessful outcomes. Stored replay receipts are inspected, not rerun by a new Rust binary here.
3. The volcano archive contains 35,290 tar members and 29,409 manifested regular files. All manifested file hashes match, with no missing entries or unmanifested extra regular files. This is integrity verification rather than mathematical validation.
4. The cryptanalysis native library builds with GCC 14.2.0 and CMake 4.4, Release configuration, CUDA and cuPQC disabled. Fifteen of seventeen configured tests pass initially; the use-case test is missing its sparse files and the coordinator test is denied local sockets. After restoring those tracked files and enabling network access for the test process, both failed tests pass. Initial and rerun logs are retained. The GPU test in this CPU-only build is not a hardware GPU measurement.
5. The paper's own check builds the PDF, checks citations and source keys, confirms figure inputs and named counts, and checks for unresolved references and overfull boxes. Rendering and page inspection assess the delivered PDF, not the truth of its cited experiments.

A.3 Requirement-to-evidence coverage

Requested assessment	Status	Evidence or remaining gap
Map crypto, cryptanalysis, autoresearcher and related projects	Complete as an accessible repository inventory	Nineteen pinned trees and current branch manifests; private or uncommitted material is not inferred
Measure six months of recorded work	Complete for the defined metadata metrics	Exact window, pins, counting rules and catalogues; not a measure of labour, money or global historical rank
Explain fields, implementations and experiments	Comprehensive synthesis with selective deep audit	Main technical chapters plus 4,075 directory records and source register; not every run independently re-executed
Show concrete progress	Complete at the stated boundaries	Held-out IC configuration changes, solver and GPU stages, complete toy pipelines, structural discovery, native validation and archive checks
Distinguish proofs from hypotheses	Complete as a scoped proof map	Central source statements and missing bridges named; fresh Lean builds and full independent semantic review remain open

Requested assessment	Status	Evidence or remaining gap
Cover isogenies and graphs	Complete for accessible summaries and the checked delivery	Exact graph/measurement populations and outcomes; private packets and larger unfinished campaigns remain explicit
Review everything in every branch and external archive	Partial	All captured branch commit metadata is indexed, while branch contents and raw experiment replay are selective; the historical 51-GB mirror and private ten-lead packets are unavailable

A.4 Before external submission

This is an anonymised research draft and retrospective, not a peer-reviewed paper or an external submission. Before submission, resolve or explicitly maintain the open-branch addendum, independently review the load-bearing source claims and proof fidelity, verify the literature and priority statements against primary publications, and rebuild from the chosen submission pins. A successful manuscript build is not clearance for a security or novelty claim.

B Pinned source register

The keys used throughout the paper resolve below to immutable repository revisions or explicitly identified external artifacts. Source hashes bind bytes; they do not establish scientific truth. Open-PR entries record their observed head and state, rather than treating a mutable PR page as a frozen experiment. Full source metadata is in `evidence/sources.json`.

[C-README] Implementation inventory and public interfaces

crypto: `README.md`. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/README.md
Revision: `4f9ea9480b17250880161859815b284a598bfb6b`.
SHA-256: `91e75171a50b8724e16f57e7ea1e17b10bc7db2534dd2bf958675e5419f14413`.

[C-RULES] Evidence, cost, identity and manuscript contracts

crypto: `AGENTS.md`. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/AGENTS.md
Revision: `4f9ea9480b17250880161859815b284a598bfb6b`.
SHA-256: `8351ab1807514d986c0a4aafdbe4e256f79c300a78d4e32ca41924d397dd78cd`.

[CA-README] Native C and Rust cryptanalysis toolkit

cryptanalysis: `README.md`. https://github.com/aburan28/cryptanalysis/blob/6e170e48882e1ba21dc90f95e6eba07f3879eabc/README.md
Revision: `6e170e48882e1ba21dc90f95e6eba07f3879eabc`.
SHA-256: `78dbc67255db0e61c037de86146f287cce064332055906fb2875a37d36c05d7b`.

[AR-README] Research orchestration and scope

crypto-autoresearcher: `README.md`. https://github.com/aburan28/crypto-autoresearcher/blob/d0f70c0a0905a5d4d2fc7c082b6dd85e43094026/README.md
Revision: `d0f70c0a0905a5d4d2fc7c082b6dd85e43094026`.
SHA-256: `d9329bec243bb599eb93e608d1e08d9dde9cae2e794e95bc2b9461ce21f9a317`.

[PRIOR] Earlier index-calculus programme synthesis

crypto: `papers/ecdlp-index-calculus-program/README.md`. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/papers/ecdlp-index-calculus-program/README.md
Revision: `4f9ea9480b17250880161859815b284a598bfb6b`.
SHA-256: `5f7d927cefca8eee068e97f307edd0aec2d04ec68a52b1b70d1503472fc8e0c6`.

[AUTOLOG] Earlier experimental work log

crypto: `research/notes/index-calculus/RESEARCH_AUTOLAB_LOG.md`. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/notes/index-calculus/RESEARCH_AUTOLAB_LOG.md
Revision: `4f9ea9480b17250880161859815b284a598bfb6b`.
SHA-256: `cb0ca2d88f5d4c08df7bd45960a71590aa8118fd4c257a838082dfe83df75d41`.

[HNP] GLV and hidden-number research

crypto: `research/notes/lattice-hnp/RESEARCH_GLV_HNP.md`. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/notes/lattice-hnp/RESEARCH_GLV_HNP.md
Revision: `4f9ea9480b17250880161859815b284a598bfb6b`.
SHA-256: `e5d35fe4140b1b6c885614fc7ad579554f865c1ff55df4f4712227d1d0e5656c`.

[LLL] Corrected floating-point LLL note

crypto: `papers/lll-degeneracy-note/README.md`. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/papers/lll-degeneracy-note/README.md
Revision: `4f9ea9480b17250880161859815b284a598bfb6b`.
SHA-256: `3ae7001f16233c0ca678c3e43bf757287bb8e81a06d098b05c733af806b9e50c`.

[MIGRATION] Historical archive migration and access boundary

crypto-autoresearcher: `docs/autolab-migration-20260802.md`. https://github.com/aburan28/crypto-autoresearcher/blob/d0f70c0a0905a5d4d2fc7c082b6dd85e43094026/docs/autolab-migration-20260802.md
Revision: `d0f70c0a0905a5d4d2fc7c082b6dd85e43094026`.
SHA-256: `77919970a23baa8c91971d1d06d3d2db0c063048e80e66df427d80d0043af48c`.

[BOUNDARY] Binary, Koblitz and prime stage ledger

crypto: docs/ic/boundary_targets.json. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/docs/ic/boundary_targets.json
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 5f8e0935711e63ae1ac0056b640fdf905ff54294d2dc5b414b5cf3174f8b8666.

[LEADER] Historical leaderboard with accounting boundaries

crypto: docs/ic/LEADERBOARD.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/docs/ic/LEADERBOARD.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 8f63ffd34d3381d7f41f8f0e8ea663c1d380899107b0e36ffb943ee7dd1e9f8b.

[TIMELINE] Typed progress timeline and superseded results

crypto: docs/ic/progress-timeline.json. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/docs/ic/progress-timeline.json
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 1702ed6cb3e473febd6d3329590df418089c0c3921cdb80f27ab7b1868f5acb3.

[FRONTIER] Controlled cross-method campaign

crypto: research/ic_frontier_campaign_20261015/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ic_frontier_campaign_20261015/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: deff5fb8da274bd7703c95a33fccc33c41529074f0a30957afff342c2c862b98.

[BAND] Campaign ratios at three native-operation weights

crypto: research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 0431d052a0ec7a0d04b12076966b0723f5a88f26d03bcc19e1a056b2fe42f5f1.

[RF-AUDIT] Root-finding source and measurement audit

crypto: research/ecbench_rootfinding_assessment_20261012/REPORT.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ecbench_rootfinding_assessment_20261012/REPORT.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 6d691efbb01c2f2fd3f77ee605805a20e625038f6480afcefcfc17653e825208.

[RF-PROOF] Affine Semaev factorization proof source

crypto: papers/rootfinding-decomposition/lean/S3Roots.lean. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/papers/rootfinding-decomposition/lean/S3Roots.lean
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 98bd81e17ca6e47491b108372bfe82fc0467ccecb0f3a31ced1c8627b73cd027.

[RF-PDP] PDP root-finding oracle family

crypto: research/pdp_root_finder_20261012/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/pdp_root_finder_20261012/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: a49c6aaac52aaababb5bdb3beeba94343f49ef1e2af6d9c76e695ee511d2db2c.

[RF-GPU] GPU root-finding implementation and emulation scope

crypto: RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: c6de3c5c37b62acf8a3325c9965e6ab01fa50b351b20f4cfed814f0f912c9dab.

[GB] Groebner engine audit

crypto: docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 0d94cfb4bf63c52dedf0ec8a6654985d839b8a58c8cda571b5113ade172121c9.

[FP-F4] Prime-field F4 and external-CAS comparison

crypto: research/f4_fp_msolve_parity_20261011/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/f4_fp_msolve_parity_20261011/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 68726577d3dbe0b4ca9c989a6987a2272b75e8d626d50b6dc531f37159ae9b92.

[F4-GPU] Actual RTX 5090 F4 matrix and oracle measurements

cryptanalysis: experiments/f4-gpu-runpod-20261006/RESULT.md. https://github.com/aburan28/cryptanalysis/blob/6e170e48882e1ba21dc90f95e6eba07f3879eabc/experiments/f4-gpu-runpod-20261006/RESULT.md
Revision: 6e170e48882e1ba21dc90f95e6eba07f3879eabc.
SHA-256: e2e439efe874d18562827cd7d5c7fae5da62b165dad6d42389db5cbec88a041e.

[F4-PANEL] Kept and reverted H100 panel experiments

cryptanalysis: experiments/f4-gpu-panel-20261007/README.md. https://github.com/aburan28/cryptanalysis/blob/6e170e48882e1ba21dc90f95e6eba07f3879eabc/experiments/f4-gpu-panel-20261007/README.md
Revision: 6e170e48882e1ba21dc90f95e6eba07f3879eabc.
SHA-256: bd2fd80daf22fbe5bf40aac2199f903fdac17a5a67ec2399fd5f58a1942702f1.

[KOBLITZ] Koblitz invariant-base programme

crypto: RESEARCH_KOBLITZ_INDEX_CALCULUS.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/RESEARCH_KOBLITZ_INDEX_CALCULUS.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 3f491d881fa51fb0ebaf67dd7bc17222a6e420f44231cdca47ad29e179622bd7.

[COMPACT] Compact signed-Frobenius orbit implementation

crypto: docs/ic/COMPACT_ORBIT_PRODUCER.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/docs/ic/COMPACT_ORBIT_PRODUCER.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 72e2667c7619adc8b04e2944cf21575ca3822fad82eb050b8b079c9df648564e.

[N83] Five-point N83 cold holdout

crypto: research/n83_e2e_holdout_20261016/RESULT.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/n83_e2e_holdout_20261016/RESULT.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 8aab2fa952e4a687aef8fbed16736fe70a8c3b125321709be9956891970a5d6c.

[N83-ROWS] Exact holdout rows and paired intervals

crypto: research/n83_e2e_holdout_20261016/ic1_rows_38955665283.json. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/n83_e2e_holdout_20261016/ic1_rows_38955665283.json
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 4c296ed1a4e145cfaeff850b0a8882c5721971f607f2be2b4119d7e210c85600.

[EQUAL] Equal-advice rho control and free-relation census

crypto: research/ic_next10_20261014/09-equal-advice/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ic_next10_20261014/09-equal-advice/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: d2a57f25d113a05f5c39d659f9da107ea382665b88cf80a0e80a3a24b4989896.

[EQUAL83] N83 time- and RSS-matched preprocessing control

crypto: research/20261016-n83-precomputed-rho-control/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/20261016-n83-precomputed-rho-control/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 5658e31e60cfc2f72c3e4638ea67a8c41e4a53db4c2f50ba63f7efcf4e3fc334.

[WIDE] Frozen unrun equal-bytes wide protocol

crypto: research/equal_advice_wide_20261018/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/equal_advice_wide_20261018/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 12e6dc7d79a3599414cb7eb65ad24494413a758b92eec34a6fa52723d817c8aa.

[ROUND1] First IC improvement round and corrections

crypto: research/ic_improve_20261014/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ic_improve_20261014/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 1f2e5e2b59f07ef732108ae7fedfa07b7277a7e9a51a8935870211fd59db7ea9.

[ROUND2] Production free relations, row harvesting and genus-3 work

crypto: research/ic_improve_20261015/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ic_improve_20261015/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: ddf55c15c883b783f741f6f38f7df53ab6d5b4ad2640a817c482c4018d5dda83.

[D4] Weighted four-variable D4 formulation

crypto: research/d4_symmetrised_s5_20261015/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/d4_symmetrised_s5_20261015/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: a92fe7c4128b4989235e5b781bd9bc19b1115e44d9835e18b3d7f7800bbdf5d8.

[PKM] Prime Kummer-base end-to-end panels

crypto: research/pkm_tower_ecbench_20261013/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/pkm_tower_ecbench_20261013/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 89baa7602439f361835f1aa146b04116239717741dfebfe0fc0c851338af8e8e.

[SMALL] Prime-coordinate smallness experiments

ecdlp-prime-field-smallness: README.md. https://github.com/aburan28/ecdlp-prime-field-smallness/blob/4f20be9b36ffd929db51744bf4661221111b538d/README.md
Revision: 4f20be9b36ffd929db51744bf4661221111b538d.
SHA-256: 89aa4bec27bbad98a6877727f23f5c3e9f744aa0d70a2ef7c9b3a263caf7a33e.

[PRIME] Prime-field leads and obligations

crypto: RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: a1da94e153af461450a6f7c4c651101016b648dc86b0f77fe6caccdf5af53f13.

[RESIDUAL] Residual-walk costs and corrected phase accounting

crypto: research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 26c47d076fc423ba0c4c476c2648f4fdd83ef3301a77ad2370b58618e98c377a.

[PARITY] Extension-field parity ledger

crypto: research/notes/index-calculus/RESEARCH_RHO_PARITY_PROGRAMME.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/notes/index-calculus/RESEARCH_RHO_PARITY_PROGRAMME.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 32b0389a7206f8b0ddb200e11cc841187692f17d9875bec0dc84f40463b6bb2c.

[QUINTIC] Quintic torsion route and corrections

crypto: research/notes/index-calculus/RESEARCH_K5_TORSION_JOUX_VITSE.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/notes/index-calculus/RESEARCH_K5_TORSION_JOUX_VITSE.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: f8cfd20690210207l1f98ad46c83099a90ece89d9ea23549163742bf677d8d55.

[COVER] Weak-class cover, decomposition and navigation costs

crypto: research/notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 190655135899ac73394f84672a75784d1caaeb8e76593d5ff612f60e3b11dd38.

[DREG] Degree-closure discriminating cells and unfinished boundary

crypto: research/dreg_closure_driver_20261016/RESULTS.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/dreg_closure_driver_20261016/RESULTS.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 4ff1e09dfcc24fed2d9fc223092d07c541b43f74012d73958359f2763891c50e.

[HEUR] Factor-base product profiles and empirical degree rule

cryptanalysis: experiments/pdp-degree-heuristics/README.md. https://github.com/aburan28/cryptanalysis/blob/6e170e48882e1ba21dc90f95e6eba07f3879eabc/experiments/pdp-degree-heuristics/README.md
Revision: 6e170e48882e1ba21dc90f95e6eba07f3879eabc.
SHA-256: 77f2df2bf9730e4336f95ab37c5c573aa77d29784f88459c4843ed74a275e149.

[TRANSFER] Transport conservation and its hypotheses

crypto: docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 6e059f01b94bb5a48fe2fa6392f943e6a588e84928e23488a907cae6353a3cc5.

[FIBRES] Isogeny-fibre mechanisms and obligations

crypto: RESEARCH_ISOGENY_FIBERS_ECDLP.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/RESEARCH_ISOGENY_FIBERS_ECDLP.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 23205b00c7cc4c25fffccfe18cef4e591c72ac9c4cfc94d4f3d8a0b199bd0893.

[CONDUCTOR] Counted conductor-gap comparison

crypto: research/ecbench_conductor_gap_20261012/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ecbench_conductor_gap_20261012/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 4195dbee55f8ec08d36eb075b9cd960b38b3e00969a9947d8ba1d27ac4d11bf2.

[FIBRE-SCALE] Fibre-aware IC scaling

crypto: research/fiber_ic_scaling_20261011/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/fiber_ic_scaling_20261011/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 24cb7122a3077bb70732f64f9950c789f20b9d04eba7c4795b1779557ad76716.

[VOLCANO] Graph construction, checkpoints and measured coverage

crypto: research/full_volcano_graphwide_ic_20261017/START_HERE.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/full_volcano_graphwide_ic_20261017/START_HERE.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 1d2c0fb28f1d083f1807cf72c1d288a243d4207b9e5a48b737f51861f147335f.

[STRUCTURAL] Structural-completeness draft and proof-scope correction

crypto: papers/structural-completeness/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/papers/structural-completeness/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 79d608ae07b911ee29dedf791444f0a11e44373e2212053897f5c108b164337f.

[ML] Learning and symbolic-discovery programme

ml-cryptanalysis: README.md. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/README.md
Revision: 9c06ddee943c1b3bf8295d824305711d792b11b0.
SHA-256: 28274e93dbe383251cb3566eb82f6b6cb7a819e6c0129bc23cd9f0d8fd30a6d4.

[ML-HARD] Hardness, controls and within-class prediction

ml-cryptanalysis: experiments/hardness-m9/report.md. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/experiments/hardness-m9/report.md
Revision: 9c06ddee943c1b3bf8295d824305711d792b11b0.
SHA-256: c7fab02ad1d5bdb68ac099be2c462088a99d02864e4d0fc64923fb65c3f046d8.

[ML-GHS] Binary GHS isogeny-class labels

ml-cryptanalysis: `experiments/ghs-descent/report.md`. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/experiments/ghs-descent/report.md
Revision: `9c06ddee943c1b3bf8295d824305711d792b11b0`.
SHA-256: `9ad428f0ffa0dec3b8a7459d4eb85dd5365301d4a7e49110bfb8eebd602a41db`.

[ML-FPN] Extension-field degree measurements

ml-cryptanalysis: `experiments/fpn-degrees/report.md`. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/experiments/fpn-degrees/report.md
Revision: `9c06ddee943c1b3bf8295d824305711d792b11b0`.
SHA-256: `e605a6e05bf18849749bc7ffdca2d46f238af73e8cc28732d4b3ade2bbc0edb2`.

[ML-SMITH] Genus-3 Smith isogenies and symbolic rules

ml-cryptanalysis: `experiments/smith-isogeny/report.md`. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/experiments/smith-isogeny/report.md
Revision: `9c06ddee943c1b3bf8295d824305711d792b11b0`.
SHA-256: `86d27c3b25f52d73f7918570f7d0beb2545226145c10fe9cf45345120167a1c8`.

[ML-TWIST] Twist-bit identity and finite certificates

ml-cryptanalysis: `experiments/twist-bit-law/report.md`. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/experiments/twist-bit-law/report.md
Revision: `9c06ddee943c1b3bf8295d824305711d792b11b0`.
SHA-256: `fc9aa584176e1fbb5e9bf1ed0011ba03e22172731d54cdafe6587b28b618e1ab`.

[ML-HUMBERT] Humbert-equation recovery and validation

ml-cryptanalysis: `experiments/humbert-surfaces/report.md`. https://github.com/aburan28/ml-cryptanalysis/blob/9c06ddee943c1b3bf8295d824305711d792b11b0/experiments/humbert-surfaces/report.md
Revision: `9c06ddee943c1b3bf8295d824305711d792b11b0`.
SHA-256: `947b6bb1b167552def4fd61c87e96bf87b2af58f419bead09255ee4615bd1c9b`.

[LEAN] Shared proof library, trust gate and missing bridges

crypto-lean: `README.md`. https://github.com/aburan28/crypto-lean/blob/148f5dc389b341401e06d76abe7783c4ccf42f2e/README.md
Revision: `148f5dc389b341401e06d76abe7783c4ccf42f2e`.
SHA-256: `ed036386d2d7b447990d06de618643d664372b086a066b12a437bf702717b564`.

[TEN] Ten open leads and imported advisory evidence

crypto-autoresearcher: `research/ten-lead-evidence/README.md`. https://github.com/aburan28/crypto-autoresearcher/blob/d0f70c0a0905a5d4d2fc7c082b6dd85e43094026/research/ten-lead-evidence/README.md
Revision: `d0f70c0a0905a5d4d2fc7c082b6dd85e43094026`.
SHA-256: `dd589d4235e1c7b3247a3ec6dbb528d784aa8838d2e88d2ca3ac733da1a86480`.

[TORSION] Odd-torsion translation statement and formal scope

crypto-autoresearcher: `research/ten-lead-evidence/torsion-proof-note.md`. https://github.com/aburan28/crypto-autoresearcher/blob/d0f70c0a0905a5d4d2fc7c082b6dd85e43094026/research/ten-lead-evidence/torsion-proof-note.md
Revision: `d0f70c0a0905a5d4d2fc7c082b6dd85e43094026`.
SHA-256: `416dedbf6832787a2ce11b6d67cfa971e162497f897123220a22c621b6b91f94`.

[RHO-GPU] Blackwell rho throughput and release provenance

rho-blackwell: `README.md`. https://github.com/aburan28/rho-blackwell/blob/5c3db3907924927f32519723e6cc793ffc03be09/README.md
Revision: `5c3db3907924927f32519723e6cc793ffc03be09`.
SHA-256: `3ccd16bf6ccbb48ac80e7333942cb16dd1e7f9680efa63e268bfb84eb9ab664c`.

[ECC-SERVICE] Distributed challenge archive and verification service

ecc-challenges: `README.md`. https://github.com/aburan28/ecc-challenges/blob/4b4d47349b53d827d2e64e6437d5147b96e8f9b9/README.md
Revision: `4b4d47349b53d827d2e64e6437d5147b96e8f9b9`.
SHA-256: `a4cb5d7f7fdcf440c7600ec5784ed62f869243bc4640ceccf86a07d60a79b57e`.

[EIGEN] Eigen-Seta implementation and conjectural security tiers

eigenx: README.md. https://github.com/aburan28/eigenx/blob/88f88a2da7bcb6652700b2250ec327c2c94b39c5/README.md
Revision: 88f88a2da7bcb6652700b2250ec327c2c94b39c5.
SHA-256: 6437256d17f24a9728920c8037f145dffb8c4dd84fdd69bbd3cc04c016795a0a.

[GHS-MAIN] Main-branch scaffold

ghs-trapdoor: README.md. https://github.com/aburan28/ghs-trapdoor/blob/93981a150972a9dec12a145eb207948b2423b488/README.md
Revision: 93981a150972a9dec12a145eb207948b2423b488.
SHA-256: c944261c6643aef65123740eeb8c2bf21ae9a863a9404f091eb297968cb499ef.

[COST] Executable generic-group cost challenge

ecdlp-cost-challenge: README.md. https://github.com/aburan28/ecdlp-cost-challenge/blob/2e215f1d503e166414f2f63d87d0f18a37852f69/README.md
Revision: 2e215f1d503e166414f2f63d87d0f18a37852f69.
SHA-256: 5defb65d03a17dd129f09a819f5e83f72f4287d9d98c29f5fcd4f54e8c2d0031.

[QUANTUM] Quantum point-addition work log and validation boundary

crypto: quantum_circuit_secp256k1/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/quantum_circuit_secp256k1/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 8ffc7fa281687b2152c59d1e95e2feb21c6df5b9b2bbe741bcbab68dcca95ad7.

[ARCHIVES] Historical cryptologic source archive

cryptologic-archives: README.md. https://github.com/aburan28/cryptologic-archives/blob/1096f83076a7baf8d9e9db4498427f196e172787/README.md
Revision: 1096f83076a7baf8d9e9db4498427f196e172787.
SHA-256: 0f1b2b806357d60c856267770873c6907f6276af9813eb6a4ee7b6d333766343.

[LEADS18] Recent desk review and its recorded corrections

crypto: research/ic_leads_review_20261018/README.md. https://github.com/aburan28/crypto/blob/4f9ea9480b17250880161859815b284a598bfb6b/research/ic_leads_review_20261018/README.md
Revision: 4f9ea9480b17250880161859815b284a598bfb6b.
SHA-256: 7297ca06d4e65d358cac9397da079eb245f87dea3521bb110c05f77f4a2a0d64.

[PR2160] research: large-memory IC runs on a 124 GB host (PKM m=3 N=21; shape G1 frozen cells) (open PR snapshot)

crypto: pull/2160. https://github.com/aburan28/crypto/pull/2160
Revision: f813285680b01ac6ae3f6d220c326e50a268e93d.
open, unmerged at read; claims are author-reported.

[PR2412] experiment: EXP-SEMAEV-93d5cc matched legacy/repaired s4_expr rerun of the archived S4 consumers (open PR snapshot)

crypto-autoresearcher: pull/2412. https://github.com/aburan28/crypto-autoresearcher/pull/2412
Revision: 5dcd02c6ef8e94d611a77792d2523e99d97528dc.
open, unmerged at read; claims are author-reported.

[PR953] curve: fix the GLV walk's fruitless cycles (−34% group ops on j=1728, −23% on j=0) (open PR snapshot)

cryptanalysis: pull/953. https://github.com/aburan28/cryptanalysis/pull/953
Revision: 15ed910396c2f12788544eb118b33c945cbfe7b2.
open, unmerged at read; claims are author-reported.

[PR2223] ecbench: prime-field PKM tower oracle; tower F4, signature F5 and an F6 analogue on prime curves (preregistered round) (open PR snapshot)

crypto: pull/2223. https://github.com/aburan28/crypto/pull/2223
Revision: 895c1c8c4f69835b15c0b943f74160d3ee1cf274.
open, unmerged at read; claims are author-reported.

[PR2236] Add fully synthetic volcano/ECDLP experiment at 30, 40, and 50 bits (open PR snapshot)

crypto: `pull/2236`. https://github.com/aburan28/crypto/pull/2236
Revision: `e4defc39201fded82270679f65eb55892babaf89`.
open, unmerged at read; claims are author-reported.

[GHS-BRANCH] GHS mechanism prototype on a separate branch

ghs-trapdoor: `README.md`. https://github.com/aburan28/ghs-trapdoor/blob/7c27311b25144c83071a18e3e638c8a2847b4126/README.md
Revision: `7c27311b25144c83071a18e3e638c8a2847b4126`.

[VOL-ARCHIVE] Downloaded 2,935-vertex checkpoint archive

https://drive.google.com/file/d/1IoPt5C-EYlo488pPVWPx-qpDHo7t1GwY/view
SHA-256: `10f63f511b17b0f0ae7af5ed7470bafcf861e672036cbe37428369e43d0f4109`.
all 29,409 manifest entries freshly checked; no solver replay.

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[1] Claus Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica*, 147(1): 75–104, 2011.

[2] Jean-Charles Faugère, Pierrick Gaudry, Louise Huot, and Guénaël Renault. Using symmetries in the index calculus for elliptic curves discrete logarithm. *Journal of Cryptology*, 27(4):595–635, 2014.

[3] Steven D. Galbraith, Robert Granger, Simon-Philipp Merz, and Christophe Petit. On index calculus algorithms for subfield curves. In *Selected Areas in Cryptography – SAC 2020*, volume 12804 of *LNCS*. Springer, 2021. Cryptology ePrint Archive, Paper 2020/1315.

[4] Pierrick Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation*, 44(12):1690–1702, 2009.

[5] Christophe Petit, Michiel Kosters, and Ange Messeng. Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields. In *Public-Key Cryptography – PKC 2016*, volume 9615 of *LNCS*, pages 3–18. Springer, 2016.

[6] John M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

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